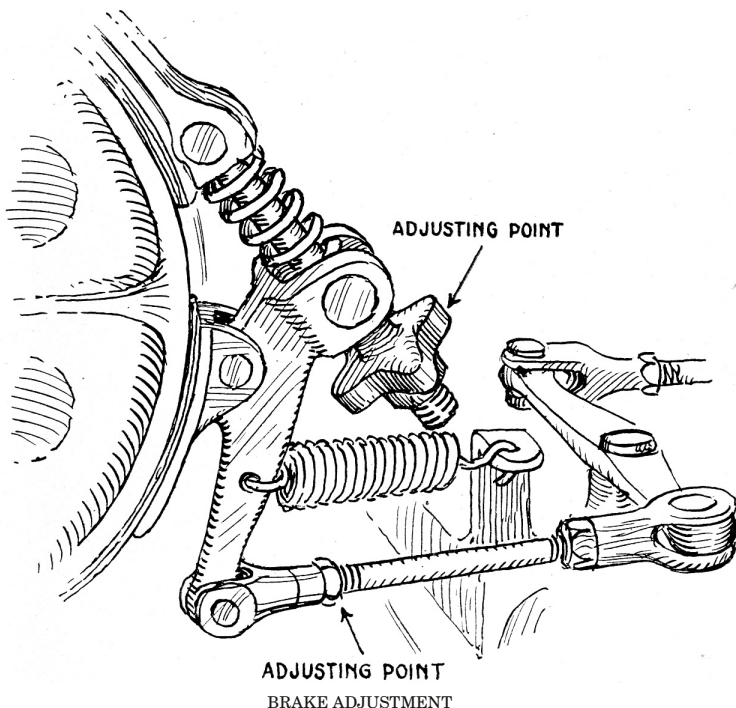


The bell crank operates against the clutch-stop spring. By regulating the length of the spring and also the position of the bell crank between the clutch-stop rod nuts, the cardan brake can be adjusted. (See Plate 95.)

#### TO ADJUST THE CLUTCH THROW-OUT

The clutch throw-out is adjusted by means of the fork ends or clevises on the clutch-control rods. The adjustment is made as nearly as possible by the use of the fork ends. The final adjustment is made by screwing up on the fork end of the rear clutch auxiliary

Plate No. 95



rod, which connects the clutch auxiliary operating lever to the clutch operating lever.

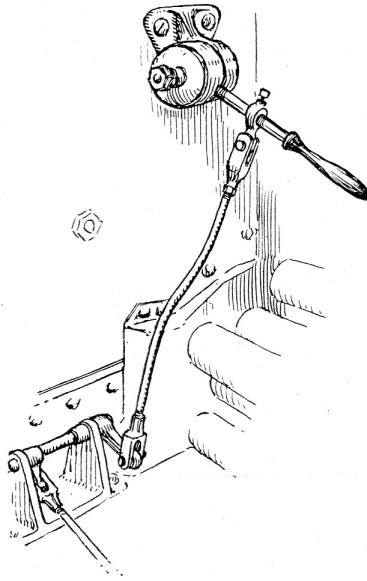
#### TO ADJUST THE HIGH-SPEED BRAKE

The first adjustments are made by screwing up on the fork ends or clevises of the control rods. The adjustment can most easily be made underneath the center ammunition storage by slacking the lock nut and turning the fork end to the desired location.

The above adjustments are what might be called permanent adjustments.

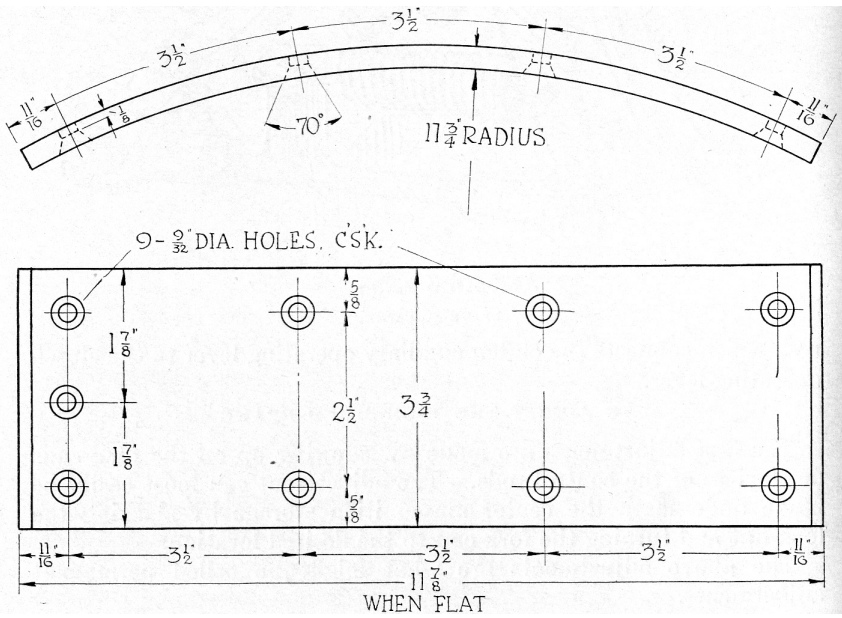
The final adjustment is made on the high-speed brake band by means of the adjusting screw (high-speed brake) which passes

Plate No. 96



SPARK OR THROTTLE LEVER

Plate No. 97



LOW GEAR BRAKE LINING

Plate No. 98

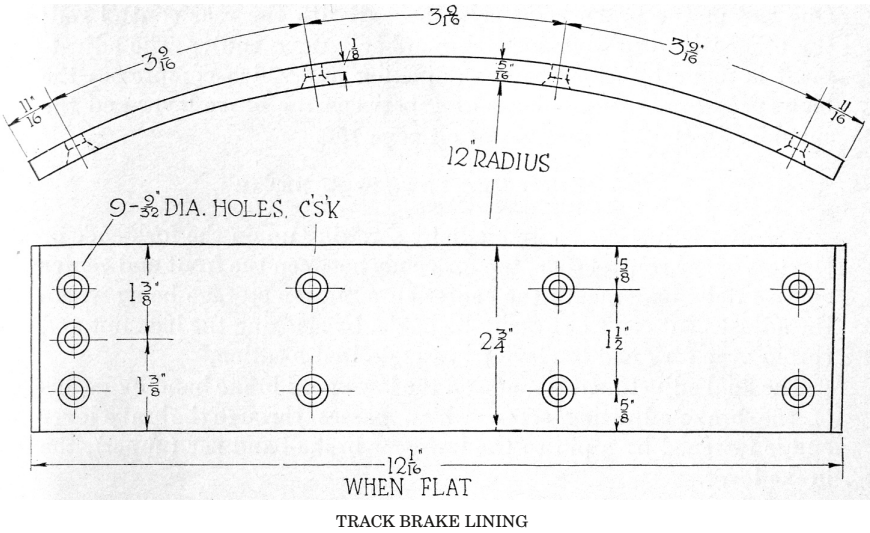
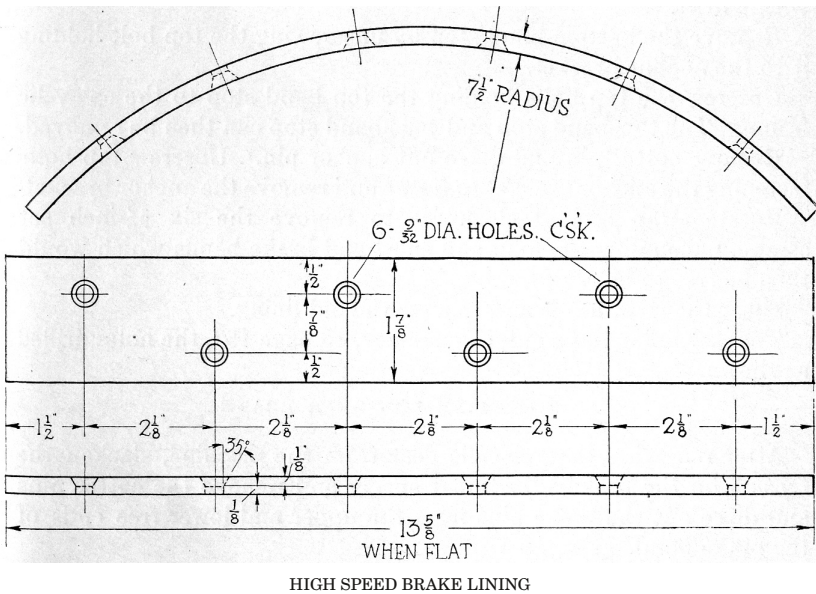


Plate No. 99



through the brake lever and connected by a pin to the high-speed brake band (short end), passing up from the under side of brake. One end of the brake lever is connected with the rear control rod; the other with the high-speed brake band (long end). The adjustment is made by screwing the adjusting nut which compresses the high-speed brake suspension spring between the brake lever and the shoulder on the adjusting screw on page 155.

#### TO ADJUST THE LOW-SPEED BRAKE

The first adjustments are made by screwing up on the fork ends or clevises of the control rods, the fork ends between the front and center control rods, underneath the center ammunition storage being where the adjustment can most easily be made, by slacking the lock nut and turning the fork end or clevis to the desired location.

The final adjustment is made on the low-speed brake band by means of the brake-adjusting screw which passes through the brake lever and is fastened by a pin to the low-gear brake-band ear (upper), the brake lever.

#### TO RELINE HIGH-SPEED BRAKE

After removing the epicyclic gear from machine, disconnect the high-speed brake and remove in the following order:

Remove cotter pins from the free end pins and drive out the pins with a drift.

Remove the bottom band stop by unscrewing the top bolt holding it to the anchor bracket.

Unscrew the tap bolts holding the top band stop to the epicyclic frame. The top band stop and backband stop can then be removed.

Remove cotter pin and drive out anchor pin. Unscrew tap bolts fastening the anchor bracket to frame and remove the anchor bracket.

Rotate strap around clockwise to remove the six  $\frac{3}{8}$ -inch set screws, uncoupling the long and short end brake bands which would then be free to be removed.

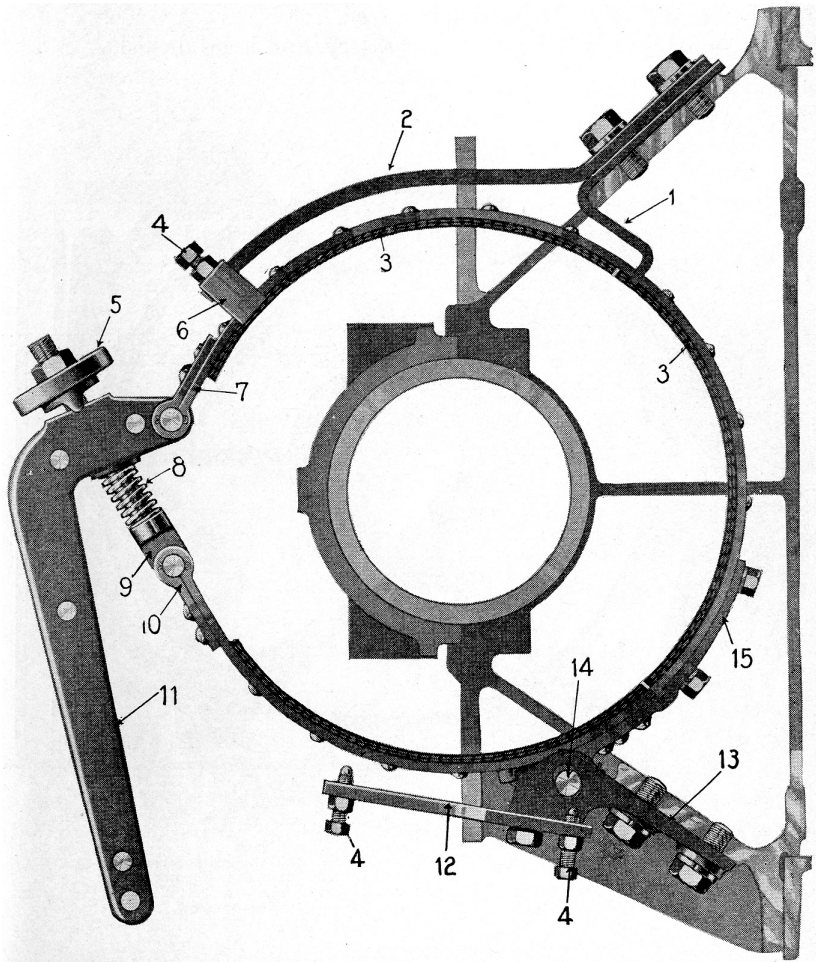
Shear the rivet heads and remove the old lining.

The new lining to be cut, if necessary, to page 153, the holes drilled as shown.

#### TO RELINE LOW-GEAR BRAKE

After removing the epicyclic gear from the machine, slacken the tension on the brake-adjustment spring and remove the cotter pins and drive out the brake pins from the upper and lower free ends of the brake band.

Remove the brake stop by unscrewing the nut of the bolt fastening it to the epicyclic frame.



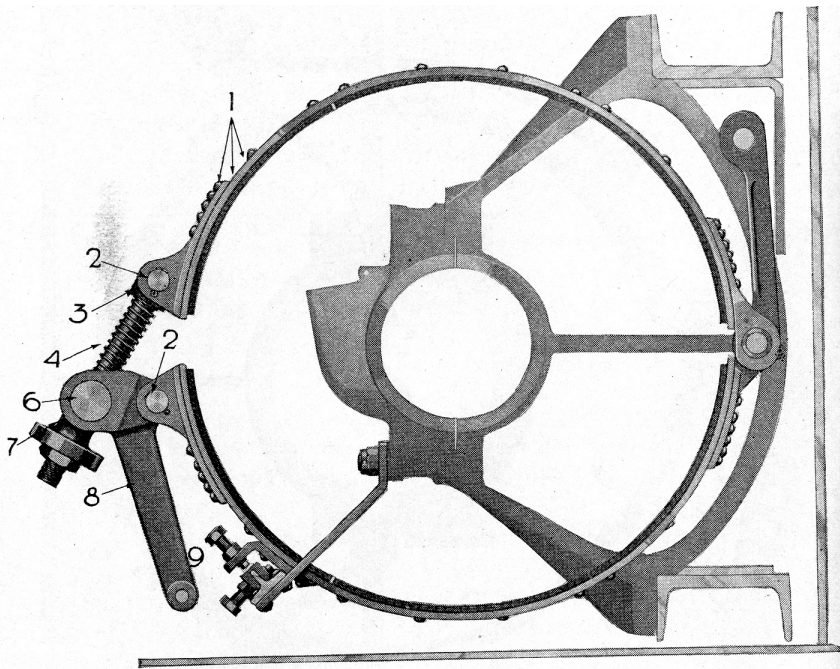
HIGH-SPEED BRAKE

| Ref.<br>No. | Part<br>No. | Name                                | Ref.<br>No. | Part<br>No. | Name                               |
|-------------|-------------|-------------------------------------|-------------|-------------|------------------------------------|
| 1           | M-398       | Backband stop, high-speed brake.    | 9           | M-358       | Adjusting screw, high-speed brake. |
| 2           | M-399       | Top-band stop, high-speed brake.    | 10          | M-360       | High-speed brake band, short end.  |
| 3           | M-364       | High-speed brake lining.            | 11          | M-355       | High-speed brake lever.            |
| 4           | M-400       | Set screw stop, high-speed brake.   | 12          | M-366       | High-speed brake bottom-band stop. |
| 5           | M-357       | Adjusting nut, high-speed brake.    | 13          | M-362       | Anchor bracket, high-speed brake.  |
| 6           | M-365       | High-speed brake-band clip.         | 14          | M-363       | Anchor pin, high-speed brake.      |
| 7           | M-359       | High-speed brake band, long end.    | 15          | M-361       | Anchor end, high-speed brake.      |
| 8           | M-369       | High-speed brake suspension spring. |             |             |                                    |

Remove the anchor pin, freeing the brake band from the brake-suspension link, and the brake band can then be rotated clockwise to remove the set screws, uncoupling the top and bottom brake bands which would then be free to be removed.

The new lining is to be cut, if necessary, and holes drilled.

Plate No. 101



TRACK BRAKE

| Ref. No. | Part No. | Name                         |
|----------|----------|------------------------------|
| 1        | M-352    | Track brake strap, complete. |
| 2        | M-336    | Brake pin, free end.         |
| 3        | M-333    | Brake-adjusting screw.       |
| 4        | M-335    | Brake-adjusting spring.      |

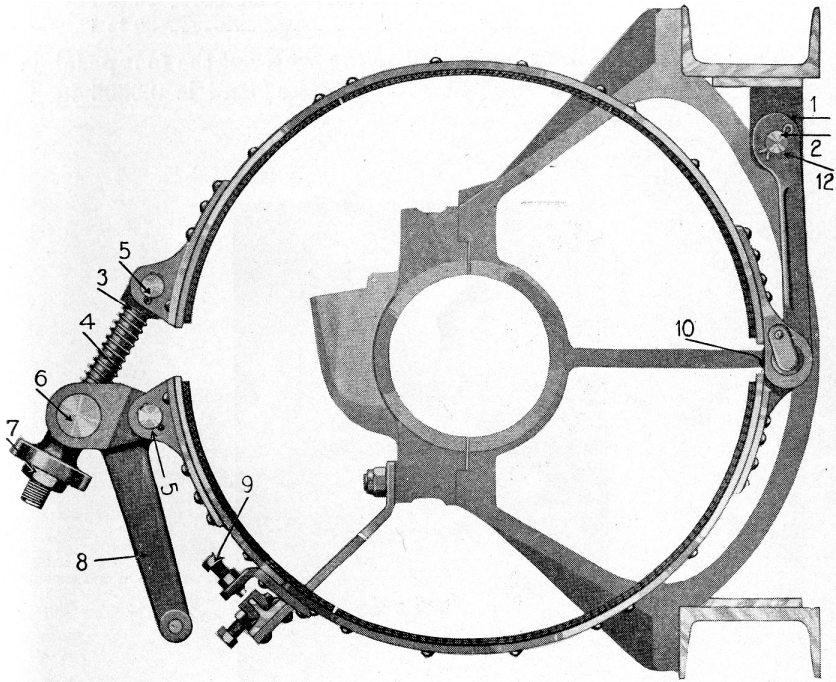
| Ref. No. | Part No. | Name                        |
|----------|----------|-----------------------------|
| 6        | M-331    | Brake-adjusting swivel pin. |
| 7        | M-332    | Brake-adjusting nut.        |
| 8        | M-330    | Brake lever.                |
| 9        | M-343    | Brake-stop screw.           |

TRACK BRAKE

The track brake is to be removed the same as low-gear brake, the new lining to be cut and drilled to sketch being fastened to the lower low-gear brake-band ear.

The adjustment is made by screwing the adjusting nut which compresses the brake-adjusting spring between the brake lever and a shoulder on the brake-adjusting screw.

[Plate No. 102]



ARRANGEMENT OF LOW-GEAR BRAKE

| Ref.<br>No. | Part<br>No. | Name                        |
|-------------|-------------|-----------------------------|
| 1           | M-338       | Brake-suspension bracket.   |
| 2           | M-337       | Brake-suspension link.      |
| 3           | M-333       | Brake-adjusting screw.      |
| 4           | M-335       | Brake-adjusting spring.     |
| 5           | M-336       | Brake pin-free end.         |
| 6           | M-331       | Brake-adjusting swivel pin. |

| Ref.<br>No. | Part<br>No. | Name                       |
|-------------|-------------|----------------------------|
| 7           | M-332       | Brake-adjusting nut.       |
| 8           | M-330       | Brake lever.               |
| 9           | M-343       | Brake-stop screw.          |
| 10          | M-348       | Low-gear brake anchor pin. |
| 12          | M-339       | Brake-suspension pin.      |

## TO ADJUST THE TRACK BRAKE

The track brake is adjusted in a similar manner to the low-speed brake, the same nomenclature and method being used.

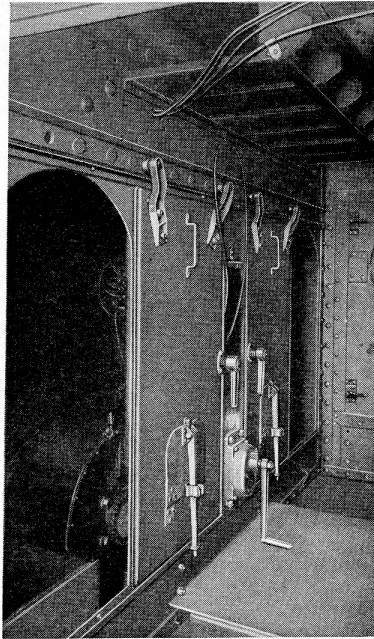
## BRAKE-CONTROL LINKAGE REDUCTION

The reduction of the brake linkage for the epicyclic gear and for the tracks is as follows:

|                        |         |
|------------------------|---------|
| Low-speed brake .....  | 52 to 1 |
| High-speed brake ..... | 26 to 1 |
| Track brake .....      | 30 to 1 |

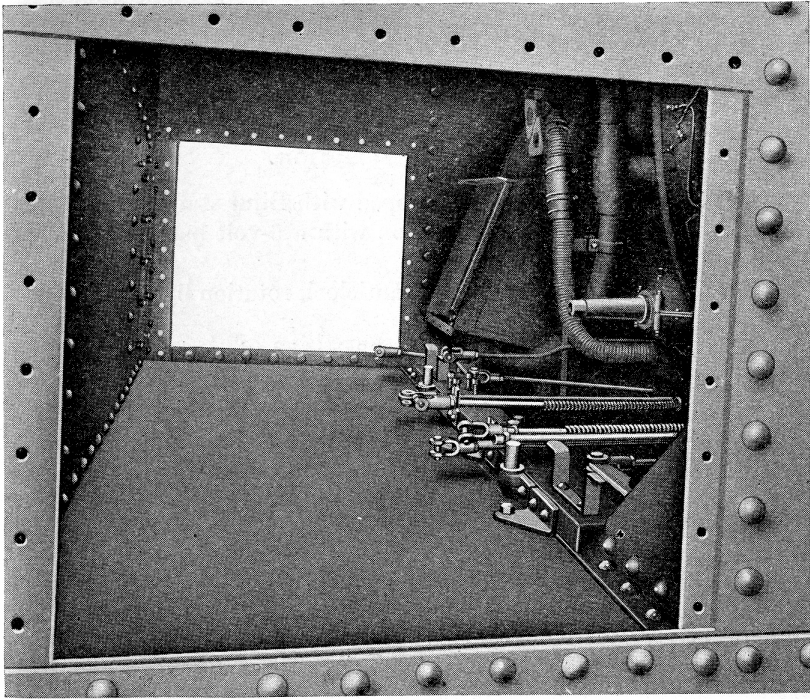
The track pedal throw, measured to the center of the foot pedal, is 11.088 inches. The movement of the brake band is 0.3696 inch. These dimensions are approximate.

Plate No. 103



BULKHEAD WITH INSTRUMENT BOARD REMOVED

Plate No. 104



CONTROL ROD ENDS WITH EPICYCLIC TRANSMISSION REMOVED FROM HULL

## Chapter IX

# ELECTRICAL EQUIPMENT

### *Outline specifications*

|                                 |                   |
|---------------------------------|-------------------|
| Starting voltage .....          | 6.                |
| Lighting voltage .....          | 6.                |
| Maximum generator voltage ..... | 6.5.              |
| Starting motor winding .....    | Series.           |
| Generator winding .....         | Shunt.            |
| Generator rotation .....        | Anticlockwise.    |
| Starting motor rotation .....   | Anticlockwise.    |
| Battery capacity .....          | 140 ampere hours. |
| Battery voltage .....           | 6.                |

## ELECTRICAL EQUIPMENT

### GENERAL DESCRIPTION

The Mark VIII machine is equipped with Bijur starting and lighting apparatus designed for operation with a 6-volt battery and consisting of:

One (1) type L-61 M-1108 anticlock rotation lighting generator;

One (1) type AD-200 M-923 anticlock rotation starting motor;

One (1) type TI-61 M-865 motor starting switch.

The generator is in operation at all times the engine is in operation.

The starting motor is in operation only during the period of starting.

### WIRING

Page 161 shows the wiring layout on the machine and covers the ignition generator (Delco make) as well as the wiring for the Bijur apparatus, heretofore mentioned.

The Bijur generator is grounded internally and only one cable leads from same, the frame of the machine acting as a common return conductor. The generator is connected direct to the battery, as no fuse is required in the generator battery circuit.

The circuit between the storage battery, starting switch, and the starting motor is made up with heavy cable, and this circuit does not make use of the machine frame as a return circuit.

### SPECIAL STARTING INSTRUCTION

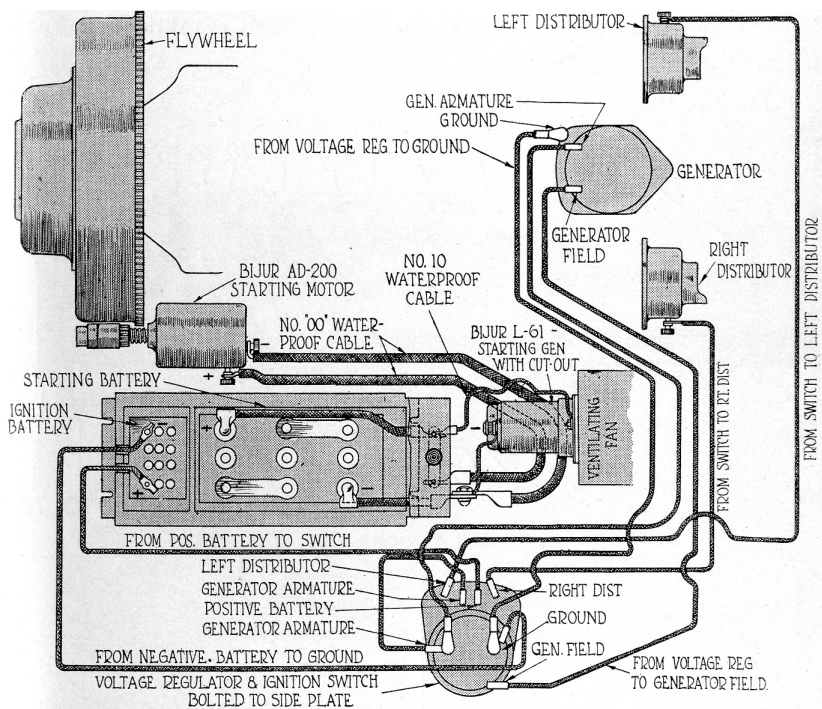
The starting motor using a 6-volt 140-ampere hour battery is not intended to crank the gas motor when cold without manual help at the hand crank.

## TO START THE GAS MOTOR

First, see that all necessary adjustments for the gas motor have been made; insert the hand crank and exert upward; pull on crank. When ready to start, signal operator at the starting switch to depress starting-switch button, at the same time giving the motor all possible assistance in turning over.

The starting switch should be depressed downward as far as it will go and held firmly in place until the gas engine begins firing. The

Plate No. 105



WIRING OF LIGHTING, STARTING, AND IGNITION

instant the gas engine begins firing, release the starting-switch button. Should the gas motor fail to start promptly after cranking it with the starting motor, do not continue to crank it, but first see that proper adjustments for starting have been made, that there is gasoline in the carburetor, and that the ignition is in working order.

Continued cranking of the gas engine with the electric starter will not damage the electrical equipment but constitutes a useless drain on the battery and should be avoided.

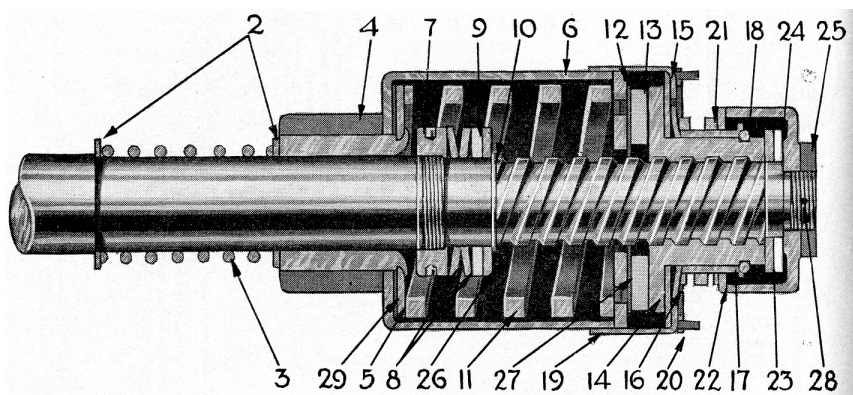
## STARTING MOTOR

### OPERATION AND DESCRIPTION

The starting motor is a series wound, four-pole machine designed for operation on a 6-volt battery, and is equipped with a Bijur automatic pinion-shifting device mounted on its armature shaft for transmitting power from the starting motor to the flywheel.

No intermediate linkage or preliminary contact resistance switch is required to bring the pinion of the starter motor into mesh with the flywheel teeth to crank the gas motor. A double-contact single-pole switch connects the starting motor directly across the battery and the instant that the switch is closed cranking is automatically accomplished.

Plate No. 106



STARTER PINION SHIFT

| Ref. No. | Name.           | Ref. No. | Name.             |
|----------|-----------------|----------|-------------------|
| 3        | Weak spring.    | 15       | Clutch plate.     |
| 4        | Pinion.         | 16       | Spring washer.    |
| 6        | Barrel.         | 17       | Sleeve.           |
| 8        | Dished washers. | 18       | Wire lock spring. |
| 9        | Thrust washer.  | 19       | Cover.            |
| 11       | Spring.         | 20       | Cover lock.       |
| 12       | Clutch plate.   | 26       | Thrust washer.    |
| 13       | Cork clutch.    | 27       | Retainer.         |
| 14       | Flanged nut.    | 28       | Threaded shaft.   |
| 15       | Pinion flange.  |          |                   |

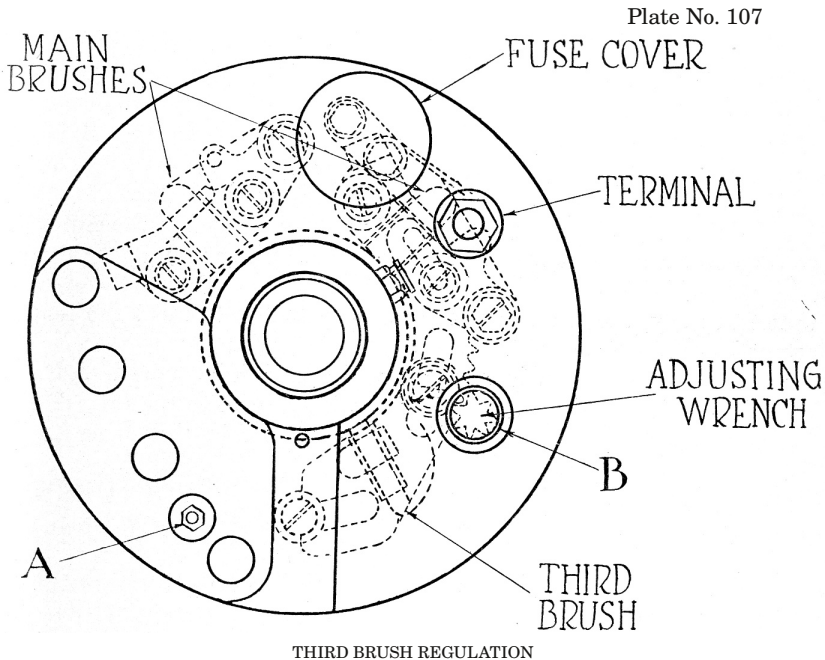
When the starting-switch pedal is depressed the starting motor to battery circuit is closed and the machine begins to operate under full power, the pinion shifting device mounted on the threaded end of the armature shaft automatically meshing with and transmitting power to the flywheel teeth, thus cranking the gas motor.

The instant that gas motor begins firing, the overriding speed at which the starter pinion is driven by the flywheel with respect to the armature shaft causes the device automatically to travel along the threaded shaft and disengage the flywheel teeth.

## STARTER PINION SHIFT

The pinion-shifting device is positive in every respect and will crank the gas motor every time the starting switch is closed. It is impossible under normal operating conditions for the starting pinion to "butt" or "jam" against the teeth of the flywheel, as the pinion will always go into mesh with the flywheel teeth for reasons which will be fully described later.

The operation and construction of the automatic gear-shifting device used in connection with the starting motor may be best understood by reference to B/P 138 and the description that is to follow:



Referring to page 162, the pinion mentioned above shown by Figure No. 4 is driven by notches in the barrel (6) registering with the pinion teeth. This barrel has 6 slots cut about  $\frac{3}{4}$  inch deep in its other end, in which the notched clutch plate (12) registers. The cork clutch (13) is held in place against (12) by means of the notched retainer (27) and its right-hand face can slip slightly against the smooth ground face of the flanged nut (14) whenever the torque exceeds the frictional hold between the cork and steel faces, this holding force being set up by the initially compressed spring (11). Power is thus transmitted in the working position from the shaft to the flanged nut (14), thence to the clutch (13), the retainer (27), clutch plate (12), barrel (6), and thus to pinion (4).

The cover (19), which fits into and over the notched barrel (6), effects closure of the shift when locked in place by spring cover lock (20).

In addition to the cork clutch there is a fixed clutch mounted on the other face of the pinion flange and shown at (15). This clutch is held against the flange by means of the clutch-spring washer (16), sleeve (17) and wire-lock spring (18). It is called "fixed" because the torque necessary to cause it to slip is constant irrespective of the position of the flanged nut.

Normal entrance is effected by depressing the starting-switch pedal, causing the shaft to rotate within the flanged nut (14), the inertia of which causes a difference in speed between the shaft and the nut with the result that the nut travels along the threaded shaft (28) and urges the entire assembly of barrel and pinion through the spring (11) into mesh. The spring (11) rests at its left-hand end against a flanged plate (5) riveted to the pinion. When the nut gets to the end of its stroke it first compresses a pair of dished washers (shown at 8) through a pair of thrust-bearing washers (9) and (26). It is the combined object of the dished washers, as well as the clutch, to take up the shock of starting. The helical spring (11) plays little, if any, role in this. Of course, when the clutch face rises enormously, that is, to say, the clutch becomes self-tightening to meet the load applied.

Coming out of mesh is caused by the overriding speed of the pinion barrel and flanged nut assembly with respect to the revolving armature shaft. The outcoming blow is cushioned by the spring (21) mounted on the flanged nut.

The purpose of the clutch plate (15) is to get sufficient initial clutch tension added to the clutch (13) to avoid slippage when a "butt" occurs.

Whenever a pinion tooth "butts," that is, strikes the corresponding end of a flywheel tooth, this interference is almost constantly overcome by the following action which takes place: The pinion is now kept from entering the flywheel teeth and the traveling nut compresses the spring (11), urging the notched barrel (6) forward along the teeth of the pinion. Finally, the clutch plate (12) strikes the washer (26) locking up, or rather tightening, the clutch (13), and in turn causes the nut (14) to turn with the shaft. The barrel (6) is therefore turned by the notched plate (12) and in turn transmits its rotative effort to the pinion (4), releasing the "butting" pinion tooth from its interference. The instant this happens the spring (11) having been greatly compressed by the previous action, snaps the offending tooth into mesh with the flywheel and normal cranking takes place.

The function of the weak springs (3) is to prevent "ticking in" of the barrel and pinion, especially when the tank is running down hill.

## TESTS TO INSURE CORRECT FUNCTIONING

The shift should always be checked for the gap between the edge of the flywheel and the pinion teeth. This distance must always be  $\frac{3}{8}$  inch. In testing the shift, run the barrel along the threads by hand and note whether there is the proper freedom of action, that is, whether the shaft is free from burrs or rust. Care should be taken that there is sufficient lubricant, oil, or graphite on the threads of the armature shaft (28) and between washers (9) and (26), washers (23) and (24).

In order to test the shift for initial torque of its clutches, a brake should be made to fit over the pinion (4) having a lever arm 1 foot in length. The armature shaft can then be held from turning by means of an open wrench on the commutator end hexagon shaft nut and force applied by hand to the end of the brake arm through a spring scale with the barrel in the "out" position. The number of foot-pounds necessary to slip the barrel in this "out" position should always be greater than  $4\frac{1}{2}$  to 5 pounds on the scale and may, however, be as much as to 12 or 15 pounds without affecting the correct functioning of the shift. If it is less than  $4\frac{1}{2}$  pounds a dangerous condition arises in case there is too tight meshing of the pinion in the flywheel. Such cases frequently require 8 to 10 pounds initial clutch torque to cause the pinion to enter and a  $4\frac{1}{2}$ -pound clutch under such conditions would slip.

## TROUBLES AND REMEDIES

(1) *Slipping clutches*.—There are two forms of clutch slip. The first is *initial* clutch slip, which has just been described, and the second is *clutch* slip, which takes place in the "in" position.

In the first case three-quarters of the initial clutch value lies in the outside or fixed clutch applied through the dished spring washer (16). It is usual to look here or in the length of the sleeve (17) for the cause of initial clutch torque weakness. In the case of clutch slipping when in cranking or "in" position this can be attributed either to a great excess of oil or lubricant on the cork face of the clutch or to a mutilated cork surface. In case the projections on back of retainer (27) shear in clutch plate (12), slippage is likely to occur, and can be remedied only by supplying a new retainer.

(2) *Sticking*.—This can happen either in the "in" or "out" position. In the "out" position it is usually caused by too tight a fit of the flanged nut on the armature-shaft threads, and can be determined as aforesaid by revolving the barrel on the threads, to feel if the mechanism is free, or it may be due to dirt on the face of the thrust washer (23) which the flanged nut strikes in this position. The remedy is this: Remove the nut (25) and thoroughly clean parts (23) and (24) as well as the back of the flanged nut. Be sure

to place graphite between washers (23) and (24) when reassembling. If the flanged nut is not perfectly free on the threaded shaft, disassemble the entire shift, beginning with spring lock (20), and "lap in" the nut on the threads, using a mixture of kerosene and emery dust. Be sure to clean the shaft and nut carefully with gasoline before reassembling. Washers (9) and (26) leave the factory with graphite between their faces and it is always well to make sure that some is put between them before reassembling, especially after cleaning the shaft with gasoline. When disassembling it is not necessary to disturb the various parts mounted on the flanged nut.

Stick in the "out" position can also be caused by the back nut (25) becoming loose. This makes the "out" stop take place between the back end of pinion (4) and shaft nut (7) instead of between the faces of (24) and (25). Sticking here occurs because the parts (4) and (7) are relatively soft, and any hammering or rubbing of the pinion (4) against the face of shaft collar (25) will roughen the butting faces, causing sticking. There is a gap between (4) and (7) in normal working of the device.

Sticking in the "in" position may be due to dirt or grit getting between the washers (9) and (26) on inside of the barrel, spring lock (10) becoming dislocated, or clutch plate (12) cutting into washer (26). Be sure that the striking face of clutch plate (12) is polished clean and that plenty of graphite and lubricant are between washers (9) and (26). In case the striking face of (12) and the inside washer (26) are cutting each other, it is best to communicate with the factory and order new parts.

(3) *Shock*.—This may be due to the flattening of washers (8), or, where too tight, gripping action of clutch (13) occurs. In the case of washer flattening, do not attempt to repair these, but replace with new ones. A very little graphite or oil applied to the cork clutch surface will usually remedy a tight clutch and reduce shock.

(4) *Precautions*.—In making repairs to shift do not replace any defective or worn parts other than those supplied directly from the Bijur factory. All the parts making up complete assembly are either made of special material or are given a special heat treatment, and the substitution of inferior or improperly treated materials will not allow the satisfactory operation of the shift.

## CONSTANT CURRENT GENERATOR

### OPERATION AND DESCRIPTION

This is a self-contained unit with no regulator or automatic cut-out requiring separate mounting and adjusting. The automatic switch for opening and closing the circuit between the storage battery and the generator is mounted inside of the generator. The regulation of the so-called constant current generator is inherent and no separate regulator is required.

The amount of current generated is, to a large extent, independent of the state of charge of the storage battery or the amount of lamp load connected, but depends primarily upon the speed at which machine is driven and the position of the regulating brush with respect to the two main brushes.

Beginning at zero speed, the voltage is of course zero, and with increasing speed the voltage increases until the armature develops 6.5 volts, at which value the shunt coil of the cut-out is sufficiently energized to cause the cut-out switch to close.

After the cut-out is closed the generator begins to deliver current to the battery or to both the battery and the light, if lights are in use. The point at which the cut-out closes (6.5 volts) corresponds to approximately 500 to 600 generator r. p. m. With increasing speed the current increases to a maximum value, at speeds of about 1,400 to 1,600 r. p. m.; at higher speeds the current decreases slightly.

The constant current generator has a single shunt winding distributed over four poles and the regulation is effected by having this winding connected between one of the main generator brushes and an auxiliary or regulating brush, and about 60° from this main brush. The maximum current generated depends upon the location of the third brush with respect to the main brush to which one side of the shunt field is connected. Moving the third, or regulating brush, in the direction of rotation of the armature increases the generator output, and in direction opposite to the rotation of armature decreases the output.

#### REGULATOR

Referring to page 163, showing the commutator end housing of the type L-61 generator, and considered for clearness to be transparent, the third brush and the two main brushes are indicated by name, the third brush adjusting hole at B, and the main switch-adjusting nut at A. Figure 165 shows all internal connections for the shunt winding, brushes, and the main switch.

Due to the fact that the voltage regulation of all third brush generators is effected by means of the reaction magnetic flux set up by the current flowing through the armature, it is necessary to introduce some protective device, such as a fuse in the field circuit, in order to prevent too great an increase in voltage across the field in case of an accidental loose or open battery connection. With the battery connected to the generator and all cable connections making good contact, the current flowing through the armature produces the necessary reaction and effects regulation holding down the voltage to a normal value.

When the generator is disconnected from the battery for any length of time, the reaction normally set up by the armature ap-

proaches a zero value, thereby causing a greatly increased e. m. f. to be developed by armature conductors between the main and third brush across which the field windings are connected. The net result under these conditions is an increased field exciting current approaching in value the normal output rating of the machine, and consequently a greatly increased terminal voltage developed. As the value of the current required for field excitation is quite small in comparison with the generator output, the field windings would not stand this overvoltage in case of an open battery connection for any length of time, and a fuse of 8 to 12 ampere capacity is therefore connected in the field circuit to protect the winding. This fuse is held in place by two phosphor-bronze clips, the entire assembly being protected by a nickel-plated cover, as shown on Diagram 164.

It is not feasible to supply lamps directly from a constant current generator with battery disconnected unless the lamps so connected consume the entire current output of the machine. If the current required to supply the lamps is less than the generator output the lamps will be burned out. For instance, if the generator output is 15 amperes and the current consumed by the lights is 7 amperes, the generator voltage will rise until the additional 8 amperes not required by the lamps will be forced through the lamp circuit. This will naturally overload the lamps and burn them out. No attempt should, therefore, be made to operate lamps direct from a constant current generator without using a storage battery in conjunction with them.

#### GENERATOR TEST

Start the engine, allow it to run at a speed sufficient to obtain a generator r. p. m. of 1,000 or more and with a test ammeter connected between the cable leading to generator and the terminal of generator, observe the output. This output should not be less than 12 or more than 16 amperes.

Before making any adjustment, test the battery with a hydrometer and carefully inspect the brushes to see that they are seated properly on the commutator and that they are free to move up and down in their respective holders. Also inspect the commutator to see that it is clean and that the mica insulation between bars is properly undercut, as high mica and dirt will prevent satisfactory operation of generator.

If the specific gravity of the battery is 1.25 or above, and the generator is delivering from 12 and 14 amperes at this speed, no adjustment is necessary.

The arrangement of the brushes and cut-out switch in the front head of the generator is shown on blue print 164 and all symbols used in the following instructions will be found on this blue print.

## ADJUSTING THE GENERATOR SWITCH OR CUT-OUT

First, see that the main switch or cut-out connects the generator to the battery at the proper voltage. The speed of generator need not be considered in making this check, but with a test voltmeter connected across the two main brushes of machine measure the voltage at the instant the test ammeter, already connected in circuit, begins to show a charge. The correct voltage at which the main switch closes is 6.5 volts, and if the voltage is above or below this value switch should be adjusted by inserting a small fiber handle socket wrench in the adjusting hole at "A" and turning the tension-spring adjusting nut until the switch closes at the correct voltage.

## ADJUSTING THE REGULATOR

The current output of the generator when driven at speeds of 1,400 to 1,600 r. p. m. should not be less than 12 or more than 15 amperes, and its output should be adjusted accordingly.

It is not necessary to remove the generator for adjusting. The third or regulating brush is so arranged that by inserting a special tooth-adjusting wrench (B-4 260-M) the adjusting hole at B and engaging the ratchet (6) the entire regulating brush assembly can be shifted in a direction required.

Before attempting to adjust, remove the cable from the generator terminal and connect a standard test ammeter between the generator and cable removed and note the value of current. If the current delivered by the generator is less than 12 or more than 16 amperes, then adjust.

Moving the regulating brush assembly in direction opposite to the rotation of the generator's armature decreases the current output and in the same direction of armature rotation increases the output.

To get the best results, the setting should be made when the generator is hot and connected to a battery having a specific gravity.

## CARE AND MAINTENANCE

In order to maintain the following:

1. Lubricated bearings.
2. Clean brushes.
3. Clean commutator.
4. Free movement of the brushes in their holders.
5. Good contact of brushes with the commutator.
6. Brush springs bearing on brushes.

1. A few drops of thin neutral oil should be put into the oilers of the generator after every two weeks. This will insure proper lubrication of the bearings and prevent freezing to the armature shaft.

2. After the generator has been run for a length of time, carbon dust will accumulate on the brushes and brush holders, which will tend to destroy their useful functions. Wear on the brushes is natural and the dust caused by same can not be avoided from collecting on the parts inside the generator. It is possible for oil from the bearings or other sources to reach the inside of the generator and mix with the carbon dust, forming a gummy paste which readily adheres to the brushes and holders. This paste can be removed by washing the brushes in gasoline and removing from the brush holders with a cloth soaked in gasoline.

3. The surface of the commutator should be free from oil and dirt. To clean the commutator, run the engine at a moderate speed, and with No. 00 sandpaper cut into thin strips remove any dirt or soil that may have collected. Do not use emery cloth or cloth. Never put lubricant of any kind on the commutator. After cleaning, be sure that all the sand particles from the sandpaper have been blown out of the machine.

4. The brushes, when replaced in their holders, should be free to move up and down.

5. Seating of the brushes can quickly be accomplished by placing a thin strip of No. 00 sandpaper under the brush with the sanded side of the paper against the commutator bearing surface of the brush and drawn from under brush against the rotation of the armature. Pressure of the brush springs is sufficient to cause the sandpaper to sear brushes and it is not necessary to apply extra pressure on the tops of brushes with the fingers.

6. See that the brush springs are away from the head casting or screws that would cause a ground, and that they bear upon tops of brushes at the center. This will insure proper pressure exerted on the brushes by the springs.

## Chapter X

### AMMUNITION AND ARMAMENT

#### OUTLINE SPECIFICATIONS

|                                              |                |
|----------------------------------------------|----------------|
| Number of six-pounder guns .....             | 2.             |
| Number of Hotchkiss machine guns .....       | 7.             |
| Six-pounder common shell .....               | 182 rounds.    |
| Six-pounder smoke shells .....               | 26 rounds.     |
| Quantity of machine-gun ammunition .....     | 21,000 rounds. |
| Number of machine-gun ammunition racks ..... | 42.            |

#### AMMUNITION AND ARMAMENT

##### GENERAL DESCRIPTION

The ammunition storage capacity of the Mark VIII machine is 21,000 rounds of machine-gun ammunition and 208 rounds of 6-pounder ammunition. Of the 208 rounds of 6-pounder ammunition, 26 rounds are smoke shells and the balance are common shells.

##### LOCATION OF 6-POUNDER SHELLS

There are five places for storing the 6-pounder ammunition, one in each sponson, one on each side of the driver, and one in the center of the fighting compartment. The top rows of 6-pounder shells in the sponsons are smoke shells. These shells go into cylindrical cases, designed to fit them, and from which they can be removed readily. The capacity of each of the sponson magazines or ammunition storage spaces are six smoke shells on each side of the sponson underneath the gun and seven smoke shells at rear of sponson, giving 13 smoke shells on each side, or 26 total.

There are 12 common shells in each sponson magazine. The magazines on each side of the driver are capable of holding 49 shells. These are common shells, giving a total of 98 for the two sides in these magazines. The munition storage in the center of the fighting compartment is capable of taking 30 shells on each side, giving a total of 60 in the center.

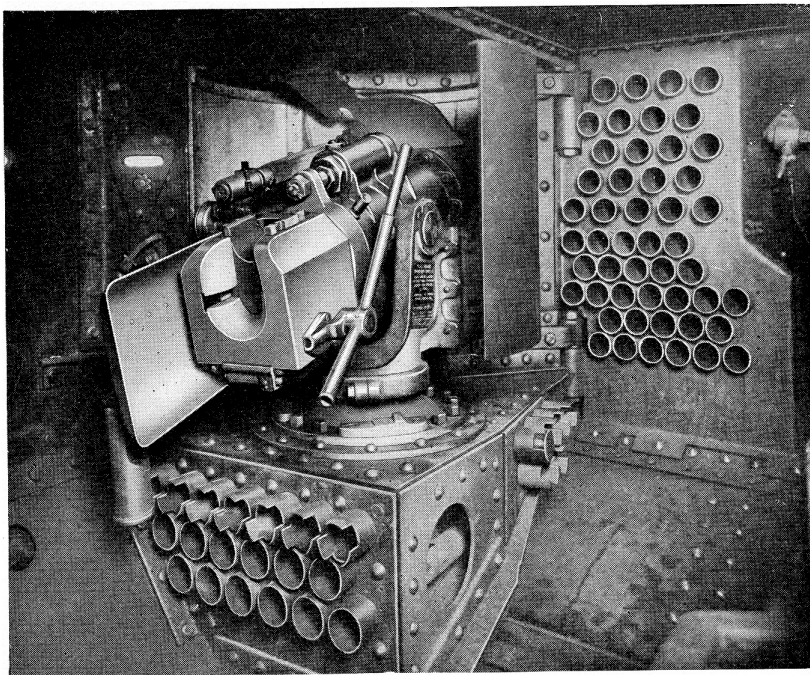
##### MACHINE-GUN AMMUNITION

The machine-gun ammunition is distributed between the ammunition storage space in the center and on the bulkhead between the engine room and the fighting compartment. The number of racks contained in the center consists of 21 on the starboard side and 17 on the port side. In the bulkhead between the engine room and the fighting compartment there are four ammunition racks.

## SPARE MACHINE-GUN BARRELS

Seven spare machine-gun barrels are carried, one being mounted in each sponson just forward of the 6-pounder shield. Two are located in each sponson, forward of the side doors. The machine guns are dismounted, and carried inside the Mark VIII machine, and are stored in the following clips for use: One on each sponson, just forward of the 6-pounder shield; two on each sponson on the rear plate of same, and one on the port side just forward of the door.

Plate No. 108



SIX-POUNDER GUN SHOWING ALSO AMMUNITION STORAGE

## SMOKE BOMBS

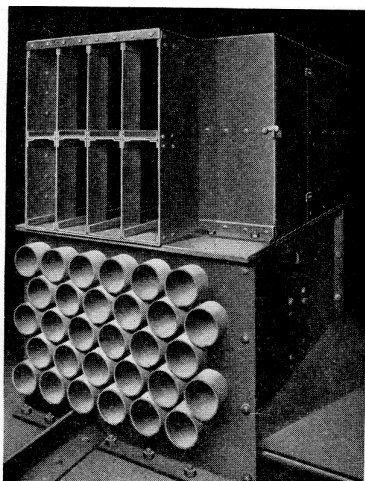
Two British .303 service rifles are stored, one on each side just aft of the doors and forward of the engine-compartment bulkhead, held by clips. Stored with these rifles are two smoke-bomb cups to be fitted to the ends of the rifles, and 80 rounds of .303 special rifle ammunition for the smoke bombs. There are also 44 rifle smoke bombs stored in a box under the port door.

## HOTCHKISS SPARE PARTS

There are two boxes of Hotchkiss spare parts, tools, and appurtenances, and each box contains the following:

- 1 fore-sight cover.
- 1 ejector.
- 1 extractor.
- 2 firing pins.
- 2 locking screws.
- 2 fore sights.
- 1 cartridge stop spring.
- 1 ejector spring.
- 2 feed springs (No. 2).
- 1 recoil spring.
- 2 sear springs.

Plate No. 109



AMMUNITION STORAGE BOX

- 1 cleaning brush.
- 1 cylinder cleaner.
- 2 hand extractors.
- 1 ejector key.
- 1 bore-cleaning rod.
- 2 dismounting wrenches.
- 1 pin box for small parts M. G.
- 1 oil can.
- 1 hammer M. G.
- 1 No. 3 M. G. punch.
- 1 mirror reflector M. G.
- 1 large screw driver M. G.
- 2 double pull throughs .303.
- 4 pieces gauze wire, pull throughs.

## SIX-POUNDER PARTS

The trays containing the 6-pounder spare parts are placed on the floor of the sponson just in the rear of the platform, one tray in each sponson. These boxes contain the following:

- 1 spare telescope (in leather case).
- 1 cleaning cloth for lenses.
- 1 cleaning rod and sponge brush.
- 1 spare sponge brush.

One set of spare parts and tools consisting of the following to be carried in two trays, one behind each gun:

- 1 spanner and screw driver, combined No. 143.
- 2 screw drivers.
- 1 oil can.
- 1 crank, complete.
- 1 crank roller.
- 2 extractors.
- 2 hand extractors.
- 2 firing hammers, complete.
- 4 firing pins, complete, with strikers.
- 1 rocking shaft.
- 2 trigger sears.
- 4 main springs.
- 4 trigger-retaining springs.
- 2 trigger sear springs.
- 1 tommy.

## LOCATION OF 6-POUNDER GUNS

The two 6-pounder guns are carried, one in each sponson. These guns swing with the sponson, and have an angle of fire from 5° right of the longitudinal line to 5° aft of the transverse line, or, in other words, in a 90° arc, moving 5° back from true longitudinal transverse. The 6-pounder guns can be pulled back within the sponson so that when the sponsons are swung inboard the guns can be withdrawn sufficiently to allow the machine to be carried on the railway.

## TO WITHDRAW THE 6-POUNDER GUNS

To pull back the 6-pounder guns for railway shipment, first unscrew the bolts fastening the sponson to the main hull, and push the sponson inboard from the outside as per directions on page 175. Unscrew the 12 bolts fastening the gun pivot plate, and the four screws fastening the training stops to the gun pedestal base plate. Draw the gun, pivot plate, and shield back as far as desired, and securely fasten to prevent sliding.

## TO REMOVE AND REPLACE THE 6-POUNDER GUNS

In removing and replacing the 6-pounder guns, remove the "above platform shell store" to give more room to work. Unscrew the 12 bolts fastening the gun pivot plate and the four screws fastening the training stops to the gun pedestal base plate. Move the gun and shield back into the fighting compartment. Remove the shield by unscrewing the six bolts holding it to the gun mount. The gun and shield can then be removed separately through the side door. To replace the gun, put the shield in through one of the side doors, and then the gun and mounting. Bolt the two together, and place on top of the gun pedestal base plate and bolt into position.

## LOCATION OF MACHINE GUNS

There are seven Hotchkiss machine guns on the Mark VIII machine, five of these being in the main turret, one front, one rear, two on each side, and one located in each of the side doors, in the hull of the machine.

## TO MOUNT MACHINE GUN

To mount the Hotchkiss machine gun in any of the machine gun posts in the Mark VIII machine, the machine gun is placed in the mount and pushed forward until the mounting shoulder falls into the recesses. The gun is then pushed forward and down. This pushes the locking lug back and the gun falls into place, being held by the locking lugs.

To remove the machine gun, push forward and up, or turn upside down so weight will help, which pushes the locking lugs in and releases the shoulder. The gun can then be removed.

## Chapter XI

### MISCELLANEOUS FITTINGS AND EQUIPMENT

#### *Ventilating system specifications*

|                                            |                                    |
|--------------------------------------------|------------------------------------|
| Fan capacity .....                         | 0.900 cubic feet at 1,400 r. p. m. |
| Fan drive .....                            | Link belt from crank shaft.        |
| Length of link belt .....                  | 61 inches.                         |
| Pitch of link belt .....                   | ½ inches.                          |
| Diameter of belt sprockets .....           | 5.101 inches.                      |
| Number of teeth in belt sprocket .....     | 32.                                |
| Center to center distance of pulleys ..... | 1 foot 9 $\frac{5}{16}$ inches.    |

#### DESCRIPTION OF VENTILATING SYSTEM

The air for ventilating the fighting compartment is drawn in through the inlet louver, above the engine, by means of a suction fan, driven from the crank shaft. The fan is a 7½-inch unit, driven by a link belt at crank-shaft speed.

The intake is drawn in through the front louver, and the air passes out through the duct to the fan, and thence through a discharge duct into the fighting compartment, through an opening beneath the starboard passageway. The fan is located under the side of the engine on the starboard side, just in the rear of the bulkhead, between the fighting compartment and the engine room. It is fastened to the floor by two angle irons.

#### CAPACITY OF FAN

The fan has a capacity of 900 cubic feet per minute at 1,400 r. p. m., against a pressure of 2 inches.

#### FAN DRIVE

The fan is driven by means of a link belt,  $\frac{5}{8}$  inch wide and ½ inch pitch. The diameters of the fan-driving sprockets are 5.101 inches, and there are 32 teeth on each, giving a 1 to 1 ratio, or, in other words, driving the fan at crank-shaft speed.

#### INLET LOUVER

The inlet louver is located in the rear of the towing post, the forward end being just where the rear end of the Mark VIII machine slopes downward. It is over the forward end of the engine compartment.

The inlet louver is made up of 34 armor-plate blades, angle-shaped, forming a 90° angle 16 mm. thick. Each of these plates is separated from the next one by a distance piece, and there are 122 of these distance pieces to the inlet louver. The total dimensions of the

louver opening are 3 feet 1¼ inches in width, and 2 feet 1.825 inches in length. The length and size of the louver opening are 6 feet 8¾ inches high, with a 2⅞-inch flange at the bottom and a ¾-inch flange at the top, these plates being made of  $\frac{5}{16}$ -inch material.

#### UNDITCHING DEVICE

The unditching device consists of two 12 by 12 skew logs, carried on the guards at the rear of the machine until they are required. When the machine is ditched the rods are chained to the tracks so that they move with the tracks. At the rear end of the turret the rods strike an angle plate, set so as to carry them by the turret at an angle. As they fall over the front of the machine they straighten themselves and lie under the track, giving additional traction to the machine, allowing it to go ahead. The logs are carried by the track back to their original position, and if the machine has become unditched they can be unjoined and stored in place.

#### EQUIPMENT OF MACHINE

The following list of equipment is tentative, and in the final edition of this handbook will probably be corrected. At the time of going to press with the preliminary edition the following items are to be carried by each Mark VIII machine. The parts marked "E." are supplied by England and the parts marked "A." are supplied by America:

##### EQUIPMENT CARRIED ON MARK VIII MACHINE

(E. Supplied by England. A. Supplied by America.)

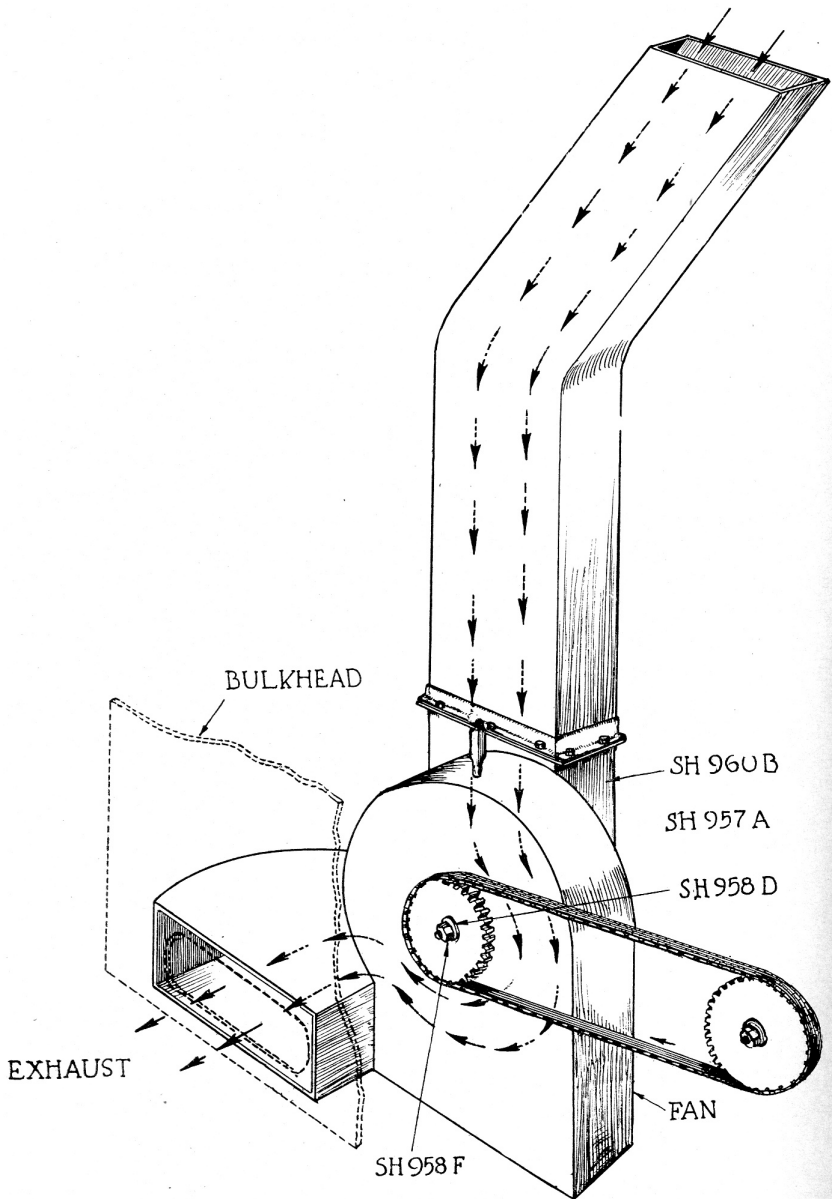
(As approved by the Tank Committee and the Anglo-American Commission, March 7, 1918.)

#### I. Guns:

##### 6-pounder (E.)—

- 26-pounder Q. F. Hotchkiss 23-caliber.
- 2 recoil mountings casement, Mark II.
- 2 sighting telescopes 15, Mark I.
- 1 spare telescope (in leather case).
- 1 cleaning cloth for lenses.
- 1 cleaning rod and sponge brush.
- 1 spare sponge brush.
- 1 set spare parts and tools consisting of the following (to be carried in two trays, one behind each gun):
- 1 spanner and screw driver combined, No. 143.
- 2 screw drivers.
- 1 oil can.
- 1 crank, complete.
- 1 crank roller.
- 2 extractors.
- 2 hand extractors.
- 2 firing hammers, complete.

Plate No. 110.



LAYOUT OF VENTILATING SYSTEM

I. *Guns*—Continued.

## 6-pounder (E.)—Continued.

- 4 firing pins, complete with strikers.
- 1 rocking shaft.
- 2 trigger sears.
- 4 main springs.
- 4 trigger-retaining springs.
- 2 trigger sear springs.
- 1 tommy.

## Machine guns (E.)—

- 7 Hotchkiss .303-inch machine guns.
- 7 spare barrels.
- 7 tripod mountings.
- 7 shackles for tripod mountings.
- 7 shoulder pieces.
- 7 cradles for 50-round ammunition belts.
- 7 cartridge-case deflectors, Marks I or II.
- 7 deflector bags.
- 44 ammunition boxes (each box to hold six 50-round belts).
- 264 50-round ammunition belts.
- 2 boxes Hotchkiss spare parts, tools, and appurtenances  
(each box containing the following) :

## Machine—

- 1 fore-sight cover.
- 1 ejector.
- 1 extractor.
- 2 firing pins.
- 2 locking screws.
- 2 fore sights.
- 1 cartridge-stop spring.
- 1 ejector spring.
- 2 feed springs (No. 2).
- 1 recoil spring.
- 2 sear springs.
- 1 cleaning brush.
- 1 cylinder cleaner.
- 2 hand extractors.
- 1 ejector key.
- 1 bore-cleaning rod.
- 2 dismounting wrenches.
- 1 pin box for small parts, M. G.
- 1 oil can.
- 1 hammer, M. G.
- 1 No. 3 M. G. punch.
- 1 mirror reflector, M. G.

I. *Guns*—Continued.

## Machine—Continued.

- 1 large screw driver, M. G.
- 2 double-pull throughs, .303.
- 4 pieces gauze wire-pull throughs.

## Smoke-bomb rifles (E.)—

- 2 service rifles, .303.
- 2 smoke-bomb cups (fitted to end of barrel).

II. *Ammunition* (E.):

- 184 rounds of Q. F., 6-pounder.
- 24 rounds of Q. F., 6-pounder case shot.
- 13,200 rounds .303 M. G.
- 80 rounds .303 special rifle for smoke bombs.
- 44 rifle smoke bombs (to be stored in box below port door).

III. *Fittings*:

## Internal—

- (E.) 1 signaling semaphore.
- (A.) 1 medical and surgical equip (carried in canvas bag and stored in bulkhead).
- (E.) 1 yardometer and odometer calibrated to read in yards.
- (E.) 1 compass (Aerial No. 259A, with electric bulb for night-work). 6 periscopes.
- (A.) 8 Pyrene fire extinguishers (3 on each door and 2 in engine room).
- (E.) 2 map boards (1 for commander and 1 for driver).
- 1 pigeon basket (to hold two birds).
- 1 broom.
- 2 electric torches.
- 1 Hellenes hand light.
- 1 inspection lamp (fitted with extension cord and plug).
- 1 hot-food container.
- 1 camp kettle.
- 2 canvas buckets.
- 1 signal gong (to be operated from outside tank).
- 1 speaking tube system.
- 4 2-gallon drinking-water cans (stored 2 above each door).
- 1 ration box (under starboard door and built into tank).
- 1 officer's locker (observation pedestal) to have folding door and lock).

## External—

- 1 tarpaulin cover.
- 1 camouflage net (to be held up by detachable posts on hull).
- 1 unditching gear (two skew logs to be stored on sponson if carried).
- 1 towing cable, 30 feet (60 tons' strength).

### III. *Fittings*—Continued.

#### External—Continued.

- 1 lock for top door (side doors to fasten from inside).
- 2 mud scrapers.
- 1 felling ax.
- 1 pickax.
- 2 shovels.
- 1 crowbar, 3 feet 6 inches.

### IV. *Tools* (to be supplied in tool box with trays and place for each tool indicated in proper tray):

#### Mechanics'—

- 1 fitter's chisel.
- 1 drift, round,  $\frac{7}{8}$ -inch.
- 1 file, half round, 6-inch.
- 1 file, round, 10-inch.
- 1 32-ounce fitter's hammer.
- 1 pair electrician's pliers.
- 2 dozen assorted split pins.
- 1 screw driver, 9-inch.
- 1 set box spanners, complete.
- 1 15-inch adjustable spanner.
- 1 small spanner.
- 1 spanner, double-ended,  $\frac{1}{8}$  by  $\frac{3}{16}$  inch.
- 1 spanner, double-ended,  $\frac{1}{4}$  by  $\frac{5}{16}$  inch.
- 1 spanner, double-ended,  $\frac{3}{8}$  by  $\frac{7}{16}$  inch.
- 1 spanner, double-ended,  $\frac{1}{2}$  by  $\frac{5}{8}$  inch.
- 1 spanner, double-ended,  $\frac{5}{8}$  by  $\frac{3}{4}$  inch.

#### *Tools* (E.):

#### Mechanics'—

- 6 spanner, double-ended,  $\frac{3}{4}$  by  $\frac{7}{8}$  inch.
- 1 spanner, double-ended,  $\frac{3}{4}$  by 1 inch.
- 1 single-ended spanner, 1-inch.
- 1 wrench pipe, 9-inch.
- 1 oil syringe.
- 4 grease guns.
- 1 roll insulating tape.
- 1 roll of 6 feet bare copper wire, No. 18, B. W. G.

#### Following supplied, but not in tool box:

- 1 wire cutter.
- 2 oil cans.

Engine (A.): Complete set of tools necessary for the general care and operation, to be shipped and carried in special tool box.

Transmission: Any special tools necessary for the adjustment and care of transmission units to be carried in a special tool box.

Track (A.): Any special tools necessary for the care and maintenance of the track units should include track pin drift, necessary punches and braces for removing grease plugs. To be carried in special tool box.

#### V. *Spares:*

##### Engine (A.)—

Spark plugs.

Generator parts.

Coil parts.

Breaker parts.

Distributor parts.

Wiring.

Fuses.

Carburetor parts.

Gaskets.

The above to be supplied in special box.

The following supplied, but not in box above.

Pump packing.

2 fan belts and spare links.

Copper tubing and connections for oiling system.

NOTE.—This list of engine spare parts is intended as a general outline. U. S. A. must advise exact spares to be furnished with engine.

Track. (from factory assembled): 1 complete road track shoe, links and pins, assembled.

Transmission (A.): Brake band linings.

Lighting system (E.):

2 spare bulbs and batteries for torches.

7 spare festoon bulbs.

2 inspection lamp bulbs.

1 Bulb and battery for Hellesen hand light.

1 Roll fuse wire.

The above to be supplied in a tin box. (Similar to ammunition box.)

#### LOCATION OF MACHINE GUNS

1 on each side in forward end on sponson.

2 on rear end of starboard sponson.

1 on rear end of port sponson.

1 on port side just forward of door.

1 on starboard side of central amm. storage opposite forward end of door.

#### SPARE BARRELS

1 on each side in forward end of sponson.

2 on starboard side just forward of side door.

2 on port side just forward of side door.

1 in main turret.

## TOWING GEAR

The towing gear consists of five brackets, one on each side of the lower part of hull just forward of the rear mud chute, one on the front and one on the back end and one on the roof.

The brackets on the sides are riveted to the longitudinal butt strap with the shackle pins vertical. The front bracket is riveted to the front sloping plate and the front floor plate with the shackle pin horizontal. The back bracket is riveted to the back and rear floor plates with shackle pin horizontal.

The side, front, and rear towing brackets have a V shackle,  $1\frac{1}{2}$  inches in diameter and  $2\frac{3}{8}$  inches between jaws, and a shackle pin  $1\frac{1}{2}$  inches in diameter which is threaded for a nut.

The roof towing bracket is bolted to the roof plate abaft of the turret by means of six  $\frac{3}{4}$ -inch bolts. It has a post  $2\frac{1}{2}$  inches in diameter and  $2\frac{1}{2}$  inches high, over which fits a steel cap 5 inches in diameter and  $1\frac{1}{8}$  inches high. The post is tapped and threaded for a 1-inch set screw, which passes through a hole in the cap and is screwed into the bracket post. To attach cable unscrew set screw and remove cap, place cable on post and replace cap.

Each machine is equipped with a steel wire cable 30 feet long by  $1\frac{1}{2}$  inches in diameter, each end of which has an eye  $4\frac{1}{2}$  inches in diameter.

## NOMENCLATURE LIST

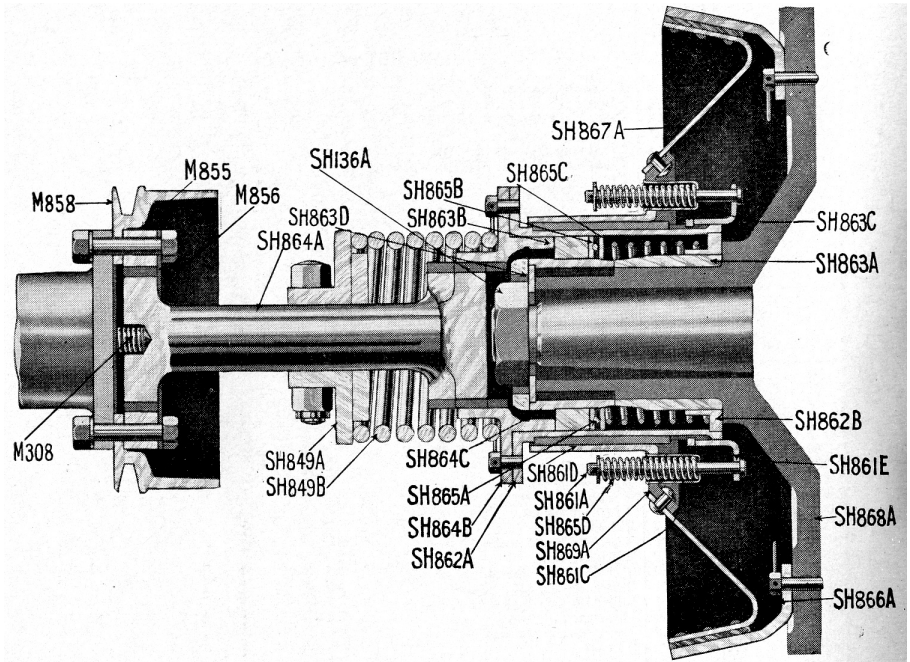
### PROPERTY CLASSIFICATION, PART I, CLASS 4, SECTION II

| Part No.                  | Number<br>per<br>machine   | Description and location                          |
|---------------------------|----------------------------|---------------------------------------------------|
| <b>AMMUNITION STORAGE</b> |                            |                                                   |
| <b>HOTCHKISS</b>          |                            |                                                   |
| M-3188 .....              | 1                          | Arrangement and detail of platform storage.       |
| M-3189 .....              | 2                          | Fork, end for detail of platform storage.         |
| M-3190 .....              | 2                          | Eyebolt for detail of platform storage.           |
| M-3191 .....              | 2                          | Adjusting nut detail of platform storage.         |
| M-3193 .....              | 1                          | Hotchkiss ammunition racks for above shell store. |
| M-3196 .....              | 1                          | Hotchkiss ammunition racks—removable section.     |
| M-3197 .....              | 1                          | Wood batten for supporting racks on bulkhead.     |
| M-3198 .....              | 1                          | Wood packing piece for racks on bulkhead.         |
| M-568 .....               | 2                          | Pin for eyebolt.                                  |
| <b>SHELL FORWARD</b>      |                            |                                                   |
| M-3063 .....              | 2                          | Shell tube plate.                                 |
| M-3064 .....              | 2                          | Shell block (upper).                              |
| M-3065 .....              | 2                          | Shell block (lower).                              |
| M-3066 .....              | 10                         | Bracket.                                          |
| M-3067 .....              | 2                          | Bracket.                                          |
| M-3068 .....              | 2                          | Bracket.                                          |
| M-3069 .....              | 2                          | Shell stop shell holder.                          |
| M-3070 .....              | 2                          | Shell stop shell holder.                          |
| M-3071 .....              | 2                          | Shell stop shell holder.                          |
| M-3072 .....              | 2                          | Shell stop shell holder.                          |
| M-3073 .....              | 2                          | Shell stop shell holder.                          |
| M-3020 .....              | 10                         | Spring case shell holder.                         |
| O-38—21192 .....          | 10                         | Spring.                                           |
| M-3023 .....              | 10                         | Plunger.                                          |
| M-3024 .....              | 10                         | Bracket.                                          |
| M-3025 .....              | 10                         | Pin.                                              |
| A-37—21191 .....          | 98                         | Detail of shell tube.                             |
| B-37—21191 .....          | 98                         | Expansion tube.                                   |
| M-2437 .....              | 4                          | Clip for periscope.                               |
| <b>SHELL REAR</b>         |                            |                                                   |
| M-3015 .....              | { (L.H.) 2}<br>{ (R.H.) 1} | Side plate platform.                              |
| M-3016 .....              | 1                          | End plate platform (rear).                        |
| M-3017 .....              | 1                          | Cover plate platform (rear).                      |
| M-3018 .....              | 1                          | End plate platform (forward).                     |
| M-3019 .....              | 2                          | Bracket strip intermediate shaft.                 |
| M-3020 .....              | 12                         | Spring case shell holder.                         |
| M-3022 .....              | 2                          | Stop-bar case shell holder.                       |
| Q-38—21192 .....          | 12                         | Spring case shell holder.                         |
| M-3023 .....              | 12                         | Plunger shell holder.                             |
| M-3024 .....              | 12                         | Bracket shell holder.                             |
| M-3025 .....              | 12                         | Pin shell holder.                                 |
| M-3026 .....              | 4                          | Bracket platform.                                 |
| A-37—21191 .....          | 60                         | Detail of shell tube.                             |
| B-37—21191 .....          | 60                         | Expansion tube.                                   |
| M-3029 .....              | 1                          | Starting handle front gearing plate.              |
| M-3030 .....              | 4                          | Stop-bar shell holder.                            |
| M-3031 .....              | 1                          | Shell block.                                      |
| M-3032 .....              | 6                          | Bracket for shell block.                          |
| <b>CARBURETOR</b>         |                            |                                                   |
| P-1 .....                 | 1                          | Body.                                             |
| P-2 .....                 | 1                          | Body gasket.                                      |
| P-3 .....                 | 2                          | Body-gasket bushing.                              |
| P-4 .....                 | 2                          | Body spring pin.                                  |
| P-5 .....                 | 2                          | Body dowel pin.                                   |
| P-25 .....                | 1                          | Float.                                            |
| P-44 .....                | 1                          | Float bolt.                                       |
| P-45 .....                | 1                          | Float-bolt nut.                                   |
| P-26 .....                | 1                          | Float chamber.                                    |
| P-27 .....                | 1                          | Float-chamber cover.                              |
| P-46 .....                | 4                          | Float-chamber cover screws.                       |
| P-47 .....                | 4                          | Float-chamber cover lock washer.                  |
| P-48 .....                | 4                          | Float-chamber stud.                               |
| P-49 .....                | 4                          | Float-chamber stud dowel pin.                     |

## NOMENCLATURE LIST—Continued

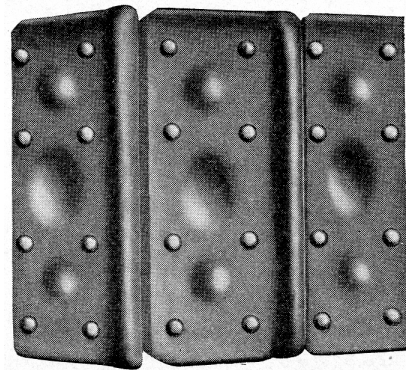
| Part No.                 | Number<br>per<br>machine | Description and location                      |
|--------------------------|--------------------------|-----------------------------------------------|
| CARBURETOR—Continued     |                          |                                               |
| P-50 .....               | 4                        | Float-chamber nut.                            |
| P-51 .....               | 4                        | Float-chamber lock washer.                    |
| P-52 .....               | 1                        | Float-chamber equalizing tube.                |
| P-53 .....               | 1                        | Float rocker shaft.                           |
| P-54 .....               | 2                        | Float rocker-shaft bearing.                   |
| P-55 .....               | 2                        | Float rocker-shaft bearing washer.            |
| P-35 .....               | 1                        | Float valve.                                  |
| P-36 .....               | 1                        | Float-valve stem.                             |
| P-37 .....               | 1                        | Float-valve collar.                           |
| P-38 .....               | 1                        | Float-valve lever                             |
| P-39 .....               | 2                        | Float-valve lever pins.                       |
| P-56 .....               | 1                        | Float-valve guide.                            |
| P-28 .....               | 1                        | Float-valve guide cap.                        |
| P-41 .....               | 1                        | Float-valve seat.                             |
| P-57 .....               | 1                        | Float-valve seat bushing.                     |
| P-42 .....               | 1                        | Float-valve seat washer.                      |
| P-58 .....               | 1                        | Float-valve stop collar.                      |
| P-250 .....              | 2                        | Air valve.                                    |
| P-251 .....              | 2                        | Air-valve cage.                               |
| P-257 .....              | 2                        | Air-valve stem.                               |
| P-255 .....              | 2                        | Air-valve stem nuts.                          |
| P-256 .....              | 2                        | Air-valve stem cotter.                        |
| P-252 .....              | 2                        | Air-valve spring.                             |
| P-258 .....              | 1                        | Air-valve locking wire.                       |
| P-150 .....              | 1                        | Horn.                                         |
| P-154 .....              | 2                        | Horn spring.                                  |
| P-600 .....              | 2                        | By-pass screws.                               |
| P-602 .....              | 2                        | By-pass plug screws.                          |
| P-300 .....              | 2                        | Throttle stems.                               |
| P-302 .....              | 2                        | Throttle blade.                               |
| P-303 .....              | 6                        | Throttle-blade screw.                         |
| P-314 .....              | 2                        | Throttle gear.                                |
| P-315 .....              | 2                        | Throttle-gear dowel.                          |
| P-316 .....              | 1                        | Throttle-gear stop screw.                     |
| P-317 .....              | 1                        | Throttle-gear lock screw.                     |
| P-550 .....              | 6                        | Gasoline-hole plug.                           |
| P-6 .....                | 2                        | Body spring pin dowels.                       |
| P-403 .....              | 2                        | Throat.                                       |
| P-404 .....              | 2                        | Throat-lock screw.                            |
| P-406 .....              | 12                       | Throat tubes.                                 |
| P-405 .....              | 2                        | Throat washers.                               |
| P-407 .....              | 2                        | Throat-lock washers.                          |
| P-650 .....              | 2                        | Idling adjusting screw.                       |
| P-653 .....              | 2                        | Idling adjusting-screw plug.                  |
| P-654 .....              | 2                        | Idling adjusting-screw bushing.               |
| P-700 .....              | 2                        | Yoke end.                                     |
| P-701 .....              | 4                        | Yoke end set screw.                           |
| P-702 .....              | 2                        | Yoke end set-screw wire.                      |
| P-703 .....              | 2                        | Yoke end bolt.                                |
| P-704 .....              | 2                        | Yoke end washer.                              |
| P-705 .....              | 2                        | Yoke end nut.                                 |
| CHAIN CASING             |                          |                                               |
| M-1583 .....             | 4                        | Angle cleats on sides.                        |
| M-1584 .....             | 2                        | Packing strip under angle cleats.             |
| M-1585 .....             | 4                        | Beading for chain casing.                     |
| M-1586 .....             | 4                        | Beading for removable cap.                    |
| M-1587 .....             | 1                        | Supporting bracket (right hand).              |
| M-1588 .....             | 1                        | Supporting bracket (left hand).               |
| M-1590 .....             | 1                        | Chain casing.                                 |
| M-1591 .....             | 1                        | Chain-casing removable cap.                   |
| M-1592 .....             | 1                        | Chain-casing angle fixing to hull back plate. |
| M-1593 .....             | 2                        | Chain-casing angle cleat on top.              |
| M-1594 .....             | 2                        | Chain-casing register plate on removable cap. |
| M-1581 .....             | 2                        | Register plate.                               |
| CLUTCH ASSEMBLY COMPOUND |                          |                                               |
| SH862A .....             | 1                        | Sliding collar.                               |
| SH861D .....             | 4                        | Keys.                                         |
| SH862B .....             | 1                        | Sliding-collar cap.                           |
| SH866A .....             | 1                        | Outer clutch drum part.                       |
| SH861A .....             | 6                        | Spring plungers.                              |
| SH861B .....             | 6                        | Washers.                                      |
| SH865D .....             | 6                        | Springs.                                      |
| SH861C .....             | 6                        | Springs retainers.                            |
| SH861E .....             | 1                        | Spring stop ring.                             |
| SH867A .....             | 1                        | Inner clutch drum.                            |
| SH869A .....             | 1                        | Clutch-drum supporting collar.                |

Plate No. 111

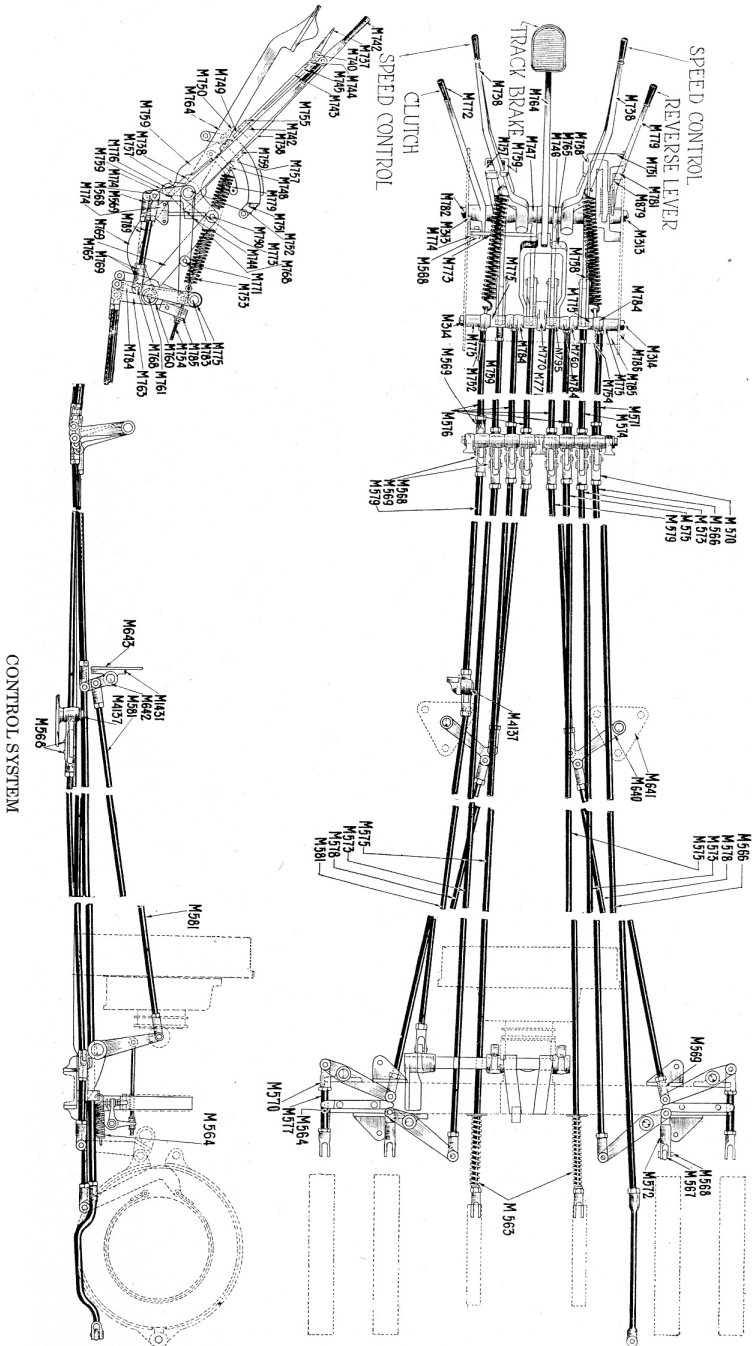


COMPOUND CLUTCH

Plate No. 112



TOP OF ROAD TRACK



## NOMENCLATURE LIST—Continued

| Part No.                           | Number<br>per<br>machine | Description and location                                                       |
|------------------------------------|--------------------------|--------------------------------------------------------------------------------|
| CLUTCH ASSEMBLY COMPOUND—Continued |                          |                                                                                |
| SH864B .....                       | 1                        | Clutch coupling.                                                               |
| SH864C .....                       | 1                        | Clutch sleeve.                                                                 |
| SH864A .....                       | 1                        | Cardon shaft.                                                                  |
| SH863A .....                       | 1                        | Clutch, front bearings.                                                        |
| SH863B .....                       | 1                        | Clutch, rear bearings.                                                         |
| SH863D .....                       | 1                        | Gear-lock washer.                                                              |
| SH863C .....                       | 1                        | Snap ring.                                                                     |
| SH855A .....                       | 2                        | Bearing race.                                                                  |
| M-855 .....                        | 1                        | Coupling box.                                                                  |
| SH865B .....                       | 1                        | Ball retainer.                                                                 |
| SH865C .....                       | 1                        | Large spring.                                                                  |
| M-856 .....                        | 2                        | Coupling cover.                                                                |
| M-858 .....                        | 1                        | Clutch-stop drum.                                                              |
| M-308 .....                        | 1                        | Cardon shaft thrust spring.                                                    |
| SH136A .....                       | 1                        | Crank-shaft nut.                                                               |
| SH849A .....                       | 2                        | Spring flange.                                                                 |
| SH849B .....                       | 1                        | Clutch spring.                                                                 |
| SH849C .....                       | 1                        | Spring-retaining ring.                                                         |
| SH849D .....                       | 1                        | Floating ring.                                                                 |
| CLUTCH SHAFT                       |                          |                                                                                |
| SH953A .....                       | 1                        | Clutch-shaft bracket (left).                                                   |
| SH953B .....                       | 2                        | Clutch auxiliary operating lever.                                              |
| SH953C .....                       | 2                        | No. 6 taper pins 2¾ inches long.                                               |
| SH953D .....                       | 1                        | Clutch auxiliary operating shaft.                                              |
| SH953D .....                       | 1                        | Clutch auxiliary rod (rear).                                                   |
| M-4163 .....                       | 3                        | ½-inch diameter cotter pins.                                                   |
| M-4163 .....                       | 1                        | Clutch lever (right).                                                          |
| M-4162 .....                       | 1                        | Clutch lever (left).                                                           |
| M-4161 .....                       | 2                        | Clutch-lever pin.                                                              |
| M-4155 .....                       | 4                        | Ball-bearing washer.                                                           |
| M-4164 .....                       | 1                        | Clutch-operating lever.                                                        |
| M-4160 .....                       | 1                        | Clutch-stop anchor plate.                                                      |
| M-4153 .....                       | 1                        | Ball-crank carrier plate.                                                      |
| M-4148 .....                       | 1                        | Clutch-shaft bracket.                                                          |
| M-4158 .....                       | 1                        | Clutch-stop band.                                                              |
| M-4159 .....                       | 1                        | Clutch-stop lining.                                                            |
| M-4150 .....                       | 1                        | Clutch shaft.                                                                  |
| M-4156 .....                       | 1                        | Clutch-stop rod eyebolt.                                                       |
| M-4172 .....                       | 5                        | Clutch-lever feather key.                                                      |
| M-4152 .....                       | 1                        | Clutch-stop rod pin.                                                           |
| M-4165 .....                       | 1                        | ½-inch split pin.                                                              |
| M-4165 .....                       | 1                        | Ball-crank pin.                                                                |
| M-4151 .....                       | 1                        | ¾-inch split pin.                                                              |
| M-4151 .....                       | 1                        | Clutch-stop rod.                                                               |
| S-673 .....                        | 1                        | Clutch-stop spring.                                                            |
| S-676 .....                        | 1                        | Clutch-stop eyebolt nut.                                                       |
| SH955A .....                       | 2                        | Clutch-stop rod nut.                                                           |
| M-4175 .....                       | 2                        | Locking set screw for clutch-lever pin.                                        |
| M-4157 .....                       | 1                        | Clutch-stop band pin.                                                          |
| M-4157 .....                       | 1                        | ½-inch split pin.                                                              |
| M-4154 .....                       | 1                        | Clutch-stop bell crank.                                                        |
| CONTROL RODS                       |                          |                                                                                |
| M-563 .....                        | 2                        | High-speed brake-rod spring.                                                   |
| M-564 .....                        | 4                        | Foot and low-speed brake-rod spring.                                           |
| M-565 .....                        | 1                        | Reversing rod (rear).                                                          |
| M-566 .....                        | 1                        | Reversing rod (center).                                                        |
| M-567 .....                        | 64                       | Spring washer (small) for brake gear.                                          |
| M-568 .....                        | 64                       | Pin for fork ends.                                                             |
| M-569 .....                        | 42                       | Fork end for control rods.                                                     |
| M-570 .....                        | 12                       | Fork end for control tube.                                                     |
| M-571 .....                        | 1                        | Reversing rod (forward).                                                       |
| M-572 .....                        | 2                        | Low-speed brake connecting tube.                                               |
| M-573 .....                        | 2                        | Low-speed brake rod (rear).                                                    |
| M-574 .....                        | 3                        | Low-speed brake rod (forward) and clutch rod (center).                         |
| M-575 .....                        | 2                        | High-speed brake (rear).                                                       |
| M-576 .....                        | 5                        | Clutch rod (forward), high-speed brake rod (center), foot-brake rod (forward). |
| M-577 .....                        | 2                        | Foot-brake connecting tube.                                                    |
| M-578 .....                        | 2                        | Foot-brake rod (rear).                                                         |
| M-579 .....                        | 2                        | Foot-brake rod (center).                                                       |
| M-581 .....                        | 1                        | Clutch rod (rear), Liberty.                                                    |

## NOMENCLATURE LIST—Continued

| Part No.            | Number<br>per<br>machine | Description and location                       |
|---------------------|--------------------------|------------------------------------------------|
| <b>CONTROL RODS</b> |                          |                                                |
| <b>ENGINE</b>       |                          |                                                |
| SH964A .....        | 2                        | Engine-control lever bracket.                  |
| SH964B .....        | 2                        | Engine-control lever hub.                      |
| SH964C .....        | 2                        | Engine-control lever friction disc.            |
| SH964D .....        | 2                        | Engine-control lever.                          |
| SH964E .....        | 2                        | Engine-control lever stud.                     |
| SH964F .....        | 4                        | Engine-control friction washer.                |
| SH964G .....        | 2                        | Engine-control lever connection.               |
| SH965A .....        | 1                        | Engine-control rocker arm bracket, long.       |
| SH965B .....        | 2                        | Engine-control rocker shaft.                   |
| SH965C .....        | 2                        | Collar for rocker shaft.                       |
| SH965D .....        | 2                        | Engine-control rocker arm bracket shaft.       |
| SH965E .....        | 4                        | Engine-control rocker arm.                     |
| SH965F .....        | 1                        | Fulcrum bracket for connecting lever.          |
| SH965G .....        | 2                        | Fulcrum bracket pin.                           |
| SH965H .....        | 2                        | Engine-control rod connection lever.           |
| SH966A .....        | 2                        | Rocker-shaft bracket for throttle control.     |
| SH966B .....        | 1                        | Rocker arm—short—for throttle control.         |
| SH966C .....        | 1                        | Rocker arm—long—for throttle control.          |
| SH966D .....        | 1                        | Rocker shaft—for throttle control.             |
| SH966E .....        | 1                        | Bell-crank lever bracket for throttle control. |
| SH966F .....        | 2                        | Bell-crank lever—spark and throttle control.   |
| SH966G .....        | 2                        | Bell-crank lever stud.                         |
| SH967A .....        | 1                        | Bell-crank lever top—bracket spark control.    |
| SH967B .....        | 1                        | Bell-crank lever for spark control.            |
| <b>ENGINE</b>       |                          |                                                |
| SH967C .....        | 1                        | Bell-crank lever stud for spark control.       |
| SH967D .....        | 1                        | Bell-crank lever bracket bottom spark control. |
| SH967E .....        | 12                       | .5-engine control clevises.                    |
| SH967F .....        | 5                        | .375-engine control clevises.                  |
| SH967G .....        | 3                        | ¾-engine control clevises.                     |
| SH968A .....        | 1                        | Engine-control rod—for throttle.               |
| SH968B .....        | 1                        | Engine-control rod—for throttle.               |
| SH968C .....        | 1                        | Engine-control rod—for throttle.               |
| SH968D .....        | 1                        | Engine-control rod—for spark control.          |
| SH968H .....        | 1                        | Engine-control rod—for spark control.          |
| SH968E .....        | 5                        | .375 clevis pins.                              |
| SH968F .....        | 12                       | .5 clevis pins.                                |
| SH968G .....        | 5                        | ¾-clevis pins.                                 |
| SH968K .....        | 1                        | .375 clevis pins.                              |
| SH969A .....        | 2                        | Engine-control rod—for spark and throttle.     |
| SH969B .....        | 1                        | Engine-control rod—for throttle.               |
| SH969C .....        | 1                        | Engine-control rod—for spark.                  |
| SH969D .....        | 2                        | Engine-control rod—vertical spark and throttle |
| SH970A .....        | 1                        | Governor spring.                               |
| SH970B .....        | 1                        | Throttle lever.                                |
| SH970D .....        | 1                        | Governor spring yoke.                          |
| SH970E .....        | 1                        | Spring—large end plate.                        |
| SH970F .....        | 1                        | Spring—small end plate.                        |
| <b>CONTROL UNIT</b> |                          |                                                |
| <b>CENTER</b>       |                          |                                                |
| M-638 .....         | 1                        | Intermediate shaft.                            |
| M-639 .....         | 3                        | Bracket for intermediate shaft.                |
| M-640 .....         | 10                       | Rocking lever for brakes.                      |
| M-641 .....         | 2                        | Fulcrum bracket for foot brake lever.          |
| M-642 .....         | 1                        | Swing link for clutch.                         |
| M-4137 .....        | 5                        | Spring washer (large) for brake gear.          |
| M-4131 .....        | 1                        | Fulcrum bracket for brake levers.              |
| <b>CONTROL UNIT</b> |                          |                                                |
| <b>FRONT</b>        |                          |                                                |
| M-764 .....         | 1                        | Brake pedal.                                   |
| M-765 .....         | 1                        | Foot-brake bridle.                             |
| M-766 .....         | 1                        | Foot-brake bridle distance.                    |
| M-767 .....         | 1                        | Foot-brake bridle pin.                         |
| M-768 .....         | 3                        | Foot-brake bridle pin nut.                     |
| M-769 .....         | 2                        | Foot-brake link.                               |
| M-770 .....         | 1                        | Foot-brake suspension link.                    |
| M-771 .....         | 8                        | Brake connecting link.                         |

## NOMENCLATURE LIST—Continued

| Part No.            | Number<br>per<br>machine | Description and location                     |
|---------------------|--------------------------|----------------------------------------------|
| <b>CONTROL UNIT</b> |                          |                                              |
| <b>FRONT</b>        |                          |                                              |
| M-738 .....         | 2                        | Control lever.                               |
| M-739 .....         | 3                        | Hand-lever trigger.                          |
| M-740 .....         | 6                        | Control-lever trigger pin.                   |
| M-741 .....         | 2                        | Control-lever pawl.                          |
| M-742 .....         | 3                        | Control-lever pawl.                          |
| M-743 .....         | 2                        | Control-lever trigger rod.                   |
| M-744 .....         | 4                        | Control-lever trigger spring.                |
| M-745 .....         | 5                        | Control-lever trigger-spring pin.            |
| M-746 .....         | 1                        | Control-lever fulcrum, starboard.            |
| M-747 .....         | 1                        | Control-lever fulcrum, port.                 |
| M-748 .....         | 2                        | Control-lever spring pin.                    |
| M-749 .....         | 2                        | Control-lever lock.                          |
| M-750 .....         | 2                        | Control-lever lock pin.                      |
| M-751 .....         | 1                        | Control gate, starboard.                     |
| M-752 .....         | 1                        | Control gate, port.                          |
| M-753 .....         | 2                        | Control-lever spring.                        |
| M-754 .....         | 2                        | Control-spring adjusting eye belt.           |
| M-755 .....         | 2                        | Control pawl stop.                           |
| M-756 .....         | 1                        | Low-speed selector lever, starboard.         |
| M-757 .....         | 1                        | Low-speed selector lever, port.              |
| M-758 .....         | 1                        | High-speed selector lever, starboard.        |
| M-759 .....         | 1                        | High-speed selector lever, port.             |
| M-760 .....         | 2                        | Low-speed brake suspension link.             |
| M-761 .....         | 2                        | Low-speed brake suspension pin.              |
| M-762 .....         | 2                        | Low-speed connecting link.                   |
| M-763 .....         | 2                        | Low-speed brake link.                        |
| M-798 .....         | 1                        | Pedal stop.                                  |
| M-799 .....         | 1                        | Clutch-lever stop.                           |
| <b>CONTROL UNIT</b> |                          |                                              |
| <b>FRONT</b>        |                          |                                              |
| M-772 .....         | 1                        | Clutch-operating lever.                      |
| M-773 .....         | 1                        | Clutch-locking bar.                          |
| M-774 .....         | 1                        | Clutch-locking bar bracket.                  |
| M-775 .....         | 4                        | Swing link distance piece.                   |
| M-776 .....         | 2                        | Foot-brake bridle pin.                       |
| M-777 .....         | 1                        | Reverse-operating lever.                     |
| M-778 .....         | 1                        | Reverse-lever trigger rod.                   |
| M-779 .....         | 1                        | Reverse-lever quadrant.                      |
| M-780 .....         | 1                        | Reverse-lever pawl.                          |
| M-781 .....         | 2                        | Reverse quadrant distance piece.             |
| M-782 .....         | 1                        | Fulcrum shaft.                               |
| M-783 .....         | 1                        | Swing-link shaft.                            |
| M-784 .....         | 4                        | Swing link.                                  |
| M-785 .....         | 1                        | Spring anchor angle.                         |
| M-786 .....         | 1                        | Seat-support plate, starboard.               |
| M-787 .....         | 1                        | Seat-support plate, port.                    |
| M-788 .....         | 2                        | Seat-support plate, angle.                   |
| M-789 .....         | 4                        | Connecting rod.                              |
| M-790 .....         | 3                        | Joint pin.                                   |
| M-791 .....         | 1                        | Seat.                                        |
| M-792 .....         | 1                        | Seat back.                                   |
| M-793 .....         | 1                        | Seat stay, forward.                          |
| M-794 .....         | 1                        | Seat stay, aft.                              |
| M-795 .....         | 1                        | Foot-brake suspension link sleeve.           |
| No numbers          | {                        | 1 Seat cushion.                              |
|                     |                          | 1 Seat-back handle.                          |
|                     |                          | 2 Seat-back handle rivet.                    |
| M-810 .....         | 4                        | Quadrant rack ferrule.                       |
| M-811 .....         | 1                        | Packing for front seat stay.                 |
| M-812 .....         | 1                        | Packing for rear seat stay.                  |
| <b>CONTROL UNIT</b> |                          |                                              |
| <b>REAR</b>         |                          |                                              |
| M-4128 .....        | 1                        | Control channel.                             |
| M-4129 .....        | 2                        | Right-hand angle cleat for control channel.  |
| M-4130 .....        | 2                        | Left-hand angle cleat for control channel.   |
| M-4131 .....        | 5                        | Fulcrum bracket for brake levers.            |
| M-4132 .....        | 2                        | Horizontal lever for foot brake.             |
| M-4133 .....        | 1                        | Horizontal lever low-speed brake, port.      |
| M-4134 .....        | 1                        | Horizontal lever low-speed brake, starboard  |
| M-4135 .....        | 2                        | High-speed brake rod spring bracket.         |
| M-4136 .....        | 2                        | Foot and low-speed brake-rod spring bracket. |
| M-4137 .....        | 4                        | Spring washer (large) for brake gear.        |

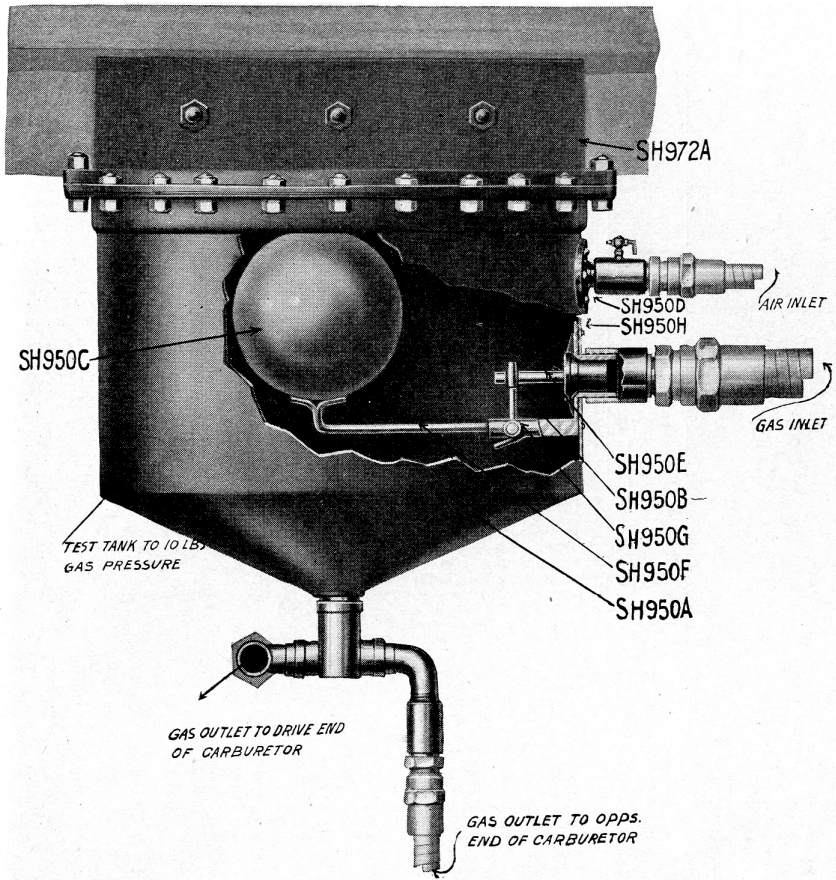
## NOMECLATURE LIST—Continued

| Part No.                     | Number<br>per<br>machine | Description and location                                                                             |
|------------------------------|--------------------------|------------------------------------------------------------------------------------------------------|
| <b>COOLING SYSTEM</b>        |                          |                                                                                                      |
| <b>AIR DUCTS</b>             |                          |                                                                                                      |
| M-1032 .....                 | 1                        | Side plate.                                                                                          |
| M-1033 .....                 | 1                        | Bottom plate.                                                                                        |
| M-1034 .....                 | 1                        | Front bottom plate.                                                                                  |
| M-1035 .....                 | 1                        | Front plate.                                                                                         |
| <b>ENGINE COOLING SYSTEM</b> |                          |                                                                                                      |
| <b>ENGINE</b>                |                          |                                                                                                      |
| 12211 .....                  | 1                        | Water-pump assembly.                                                                                 |
| 8079 .....                   | 1                        | Water-pump shaft with gear.                                                                          |
| 8056 .....                   | 1                        | Water-pump shaft ball bearing.                                                                       |
| 8069 .....                   | 1                        | Water-pump shaft bearing retainer.                                                                   |
| 8213 .....                   | 2                        | Water-pump packing.                                                                                  |
| 8059 .....                   | 2                        | Water-pump shaft gland.                                                                              |
| 8058 .....                   | 1                        | Water-pump shaft gland engine.                                                                       |
| 12071 .....                  | 1                        | Water-pump body.                                                                                     |
| 12075 .....                  | 2                        | Water-pump outlet connection (cast with body) (tube 1½ inches O. D. by 0.065 wall by 1½ inches).     |
| 8133 .....                   | 1                        | Water-pump body drain (plug ¾ inch by 18 thd., hex. head).                                           |
| 158 .....                    | 1                        | Water-pump body drain plug gasket (¾ inch annular).                                                  |
| 131 .....                    | 8                        | Water-pump cover stud (¼ inch–28 by 1½ inches, standard) (¼ inch between nut and surface).           |
| 159 .....                    | 1                        | Water-pump shaft key (⅝ inch by ½ by ¾ inch)                                                         |
| 12517 .....                  | 1                        | Water-pump impeller.                                                                                 |
| 8214 .....                   | 1                        | Water-pump shaft nut.                                                                                |
| 212 .....                    | 1                        | Water-pump shaft cotter (rustproof).                                                                 |
| 8215 .....                   | 1                        | Water-pump cover gasket.                                                                             |
| 12435 .....                  | 1                        | Water-pump cover.                                                                                    |
| 12081 .....                  | 1                        | Water-pump inlet connection (cast with cover) (tube 2 inches O. D. by 0.065 wall by 1½ inches).      |
| 111 .....                    | 8                        | Water-pump cover stud washer (¼ inch).                                                               |
| 101 .....                    | 8                        | Water-pump cover stud nut (¾ inch by 28 thd., standard).                                             |
| 8510 .....                   | x                        | Water-pump shaft bearing retainer shim.                                                              |
| 8345 .....                   | 1                        | Water-pump shaft bearing retainer gasket.                                                            |
| 8077 .....                   | 1                        | Water-pump bevel driver.                                                                             |
| 8239 .....                   | 1                        | Water-pump bevel-driver bushing and housing assembly.                                                |
| 8062 .....                   | 2                        | Water-pump bevel-driver bushing.                                                                     |
| 8065 .....                   | 1                        | Water-pump bevel-driver bushing housing (front).                                                     |
| 8064 .....                   | 1                        | Water-pump bevel-driver bushing housing (rear).                                                      |
| 174 .....                    | 1                        | Water-pump bevel-driver bushing dowel (⅝ inch by ¾ inch).                                            |
| 154 .....                    | 1                        | Water-pump bevel-driver bushing housing (rear) screw (¾ inch–16 by 3¼ inches).                       |
| 113 .....                    | 1                        | Water-pump bevel-driver bushing housing (rear) screw washer (¾ inch).                                |
| 177 .....                    | 1                        | Water-pump bevel-driver bushing housing (rear) screw lock wire (W. and M. Ga. No. 18 (0.0475) by 6). |
| 12123 .....                  | 1                        | Cylinder water-inlet manifold.                                                                       |
| 12096 .....                  | 1                        | Cylinder water-inlet manifold extension (right).                                                     |
| 12097 .....                  | 1                        | Cylinder water-inlet manifold extension (left).                                                      |
| 12128 .....                  | 4                        | Cylinder water-inlet manifold extension hose (1½ inches I. D. by 1½ inches O. D. by 1¾ inches long). |
| 8453 .....                   | 8                        | Cylinder water-inlet manifold extension hose clamp assembly.                                         |
| 8152 .....                   | 12                       | Cylinder-water hose (inlet); (see cylinder parts list also).                                         |
| 8450 .....                   | 24                       | Cylinder water-inlet hose clamp.                                                                     |
| 8426 .....                   | 1                        | Water-pump bevel-driver and housing assembly.                                                        |
| 116 .....                    | 2                        | Water-pump bevel-driver housing clamp bolt.                                                          |
| 174 .....                    | 1                        | Water-pump bevel-driver bushing dowel.                                                               |
| 111 .....                    | 4                        | Water-pump bevel-driver housing clamp bolt washer.                                                   |
| 8133 .....                   | 1                        | Water-pump body drain plug.                                                                          |
| 101 .....                    | 2                        | Water-pump bevel-driver housing clamp bolt nut.                                                      |
| 106 .....                    | 2                        | Water-pump bevel-driver housing clamp bolt cotter.                                                   |
| <b>COOLING SYSTEM</b>        |                          |                                                                                                      |
| <b>RADIATOR</b>              |                          |                                                                                                      |
| M-878 .....                  | 1                        | Top header.                                                                                          |
| M-879 .....                  | 1                        | Bottom header.                                                                                       |
| M-880 .....                  | 1                        | Stiffening plate for drain-pipe connection.                                                          |
| M-881 .....                  | 2                        | Cover plate for tube plate.                                                                          |
| SH607C .....                 | 606                      | Radiator tubes.                                                                                      |
| M-886 .....                  | 1                        | Stiffening plate for water connections.                                                              |
| M-887 .....                  | 1                        | Stiffening plate for steam-pipe connection.                                                          |
| M-888 .....                  | 2                        | Gasket for header.                                                                                   |
| M-889 .....                  | 1                        | Swivel pin.                                                                                          |
| M-891 .....                  | 2                        | Front support bracket.                                                                               |
| M-892 .....                  | 1                        | Rear support bracket.                                                                                |
| M-897 .....                  | 1                        | Bottom stiffening plate for steam-pipe connection.                                                   |
| M-898 .....                  | 1                        | Front angle bracket for radiator support.                                                            |
| M-899 .....                  | 1                        | Rear angle bracket for radiator support.                                                             |
| SH606A .....                 | 1                        | Tube plate (right).                                                                                  |

## NOMENCLATURE LIST—Continued

| Part No.                 | Number<br>per<br>machine | Description and location                                                   |
|--------------------------|--------------------------|----------------------------------------------------------------------------|
| COOLING SYSTEM—Continued |                          |                                                                            |
| RADIATOR—continued       |                          |                                                                            |
| SH606B .....             | 1                        | Tube plate (left).                                                         |
| SH606C .....             | 1                        | Reinforcing strip (right).                                                 |
| SH606D .....             | 1                        | Reinforcing strip (left).                                                  |
| SH607A .....             | 1                        | Middle tank gasket.                                                        |
| SH607B .....             | 2                        | Middle tank band iron.                                                     |
| SH611A .....             | 1                        | Middle tank plate (right).                                                 |
| SH611B .....             | 1                        | Middle tank plate (left).                                                  |
| SH612A .....             | 1                        | Side plate (left).                                                         |
| SH612B .....             | 1                        | Side plate (right).                                                        |
| RADIATOR SUPPORTS        |                          |                                                                            |
| M-889 .....              | 1                        | Swivel pin for radiator support (rear).                                    |
| M-891 .....              | 2                        | Radiator support bracket (front).                                          |
| M-892 .....              | 1                        | Radiator support bracket (rear).                                           |
| WATER TANK               |                          |                                                                            |
| M-1193 .....             | 1                        | Water tank.                                                                |
| M-1194 .....             | 1                        | Flange for filler pipe.                                                    |
| M-1195 .....             | 1                        | Flange for radiator pipe.                                                  |
| M-1196 .....             | 1                        | Stiffening plate for flow pipe.                                            |
| M-1197 .....             | 1                        | Filler pipe.                                                               |
| BA109—21149 .....        | 1                        | Cap for filler pipe.                                                       |
| WATER AND VAPOR PIPING   |                          |                                                                            |
| M-1228 .....             | 1                        | Water-outlet pipe from engine.                                             |
| M-1229 .....             | 1                        | Radiator inlet pipe.                                                       |
| M-1230 .....             | 1                        | Flanged socket for radiator inlet pipe.                                    |
| M-1231 .....             | 1                        | Water-tank outlet connection.                                              |
| SH975C .....             | 3                        | Radiator outlet coupling.                                                  |
| M-1233 .....             | 1                        | Radiator outlet band.                                                      |
| SH975B .....             | 1                        | Radiator outlet pipe.                                                      |
| SH975A .....             | 1                        | Radiator outlet tee.                                                       |
| M-1236 .....             | 1                        | Pump suction band.                                                         |
| SH976C .....             | 2                        | Clip for water pipe.                                                       |
| M-1238 .....             | 1                        | Radiator drain pipe.                                                       |
| M-1239 .....             | 1                        | Plate bracket for radiator drain cock.                                     |
|                          | 1                        | ½-inch bore cock inlet tapped ½-inch pipe thread outlet to take hose pipe. |
|                          | 1                        | ¾-inch by 2-inch bolt for plate bracket.                                   |
|                          | 1                        | ½-inch bore copper pipe, 16-gage, 7 feet 8 inches long.                    |
|                          | 2                        | Nipples for copper pipe, 16-gage, 7 feet 6 inches long.                    |
|                          | 2                        | Union nuts for copper pipe, 16-gage, 7 feet 6 inches long.                 |
|                          | 2                        | Union with one end screwed ¾-inch pipe thread.                             |
|                          | 1                        | ¾-inch W. I. knee with nipple.                                             |
| M-1240 .....             | 1                        | Pipe joint for water tank, outlet.                                         |
| M-1241 .....             | 2                        | Pipe joint for radiator inlet and outlet.                                  |
| M-1242 .....             | 1                        | Water-outlet pipe from engine.                                             |
| SH199C .....             | 1                        | Vapor pipe from engine-water outlet.                                       |
|                          | 2                        | Flexible coupling 2½ inches long (for 1-inch W. I. piping).                |
|                          | 4                        | Clips for coupling 2½ inches long (for 1-inch W. I. piping).               |
|                          | 1                        | 1-inch Whit. Gas. plug (on M 1229).                                        |
|                          | 1                        | Vapor pipe connecting piece.                                               |
| M-1244 .....             | 1                        | Support bracket (for engine water outlet and vapor pipe).                  |
| M-1245 .....             | 1                        | Clip for vapor pipe on engine.                                             |
|                          | 2                        | ⅝-inch and ⅜-inch bolt (with Grover washer).                               |
|                          | 1                        | ½-inch by ½-inch set screw (with grover washer).                           |
| SH199B .....             | 2                        | Hose connection (short).                                                   |
| SH976B .....             | 1                        | Water pipe under engine.                                                   |
| SH975E .....             | 1                        | Water-pipe coupling.                                                       |
| SH976A .....             | 2                        | Water-pipe clip.                                                           |
| SH974A .....             | 1                        | Pipe connection to pump.                                                   |
| SH975D .....             | 1                        | Piping outlet coupling.                                                    |
| DRIVING SPROCKET CHAIN   |                          |                                                                            |
| SH40AB .....             | 50                       | Outside side plate.                                                        |
| SH40AC .....             | 50                       | Inside side plate.                                                         |
| SH40AD .....             | 50                       | Collar bushing.                                                            |
| SH40AE .....             | 50                       | Collar-bushing pin.                                                        |
| ENGINE                   |                          |                                                                            |
| CAM-SHAFT HOUSING        |                          |                                                                            |
| 12104 .....              | 1                        | Cam shaft, right.                                                          |
| 12105 .....              | 1                        | Cam shaft, left.                                                           |
| 8100 .....               | 2                        | Cam-shaft plug (machined in end of each cam shaft).                        |
| 8400 .....               | 2                        | Cam-shaft bearing, No. 1 assembly.                                         |
| 8111 .....               | 2                        | Cam-shaft bearing, upper half, No. 1.                                      |
| 8112 .....               | 2                        | Cam-shaft bearing, lower half No. 1.                                       |

Plate No. 114



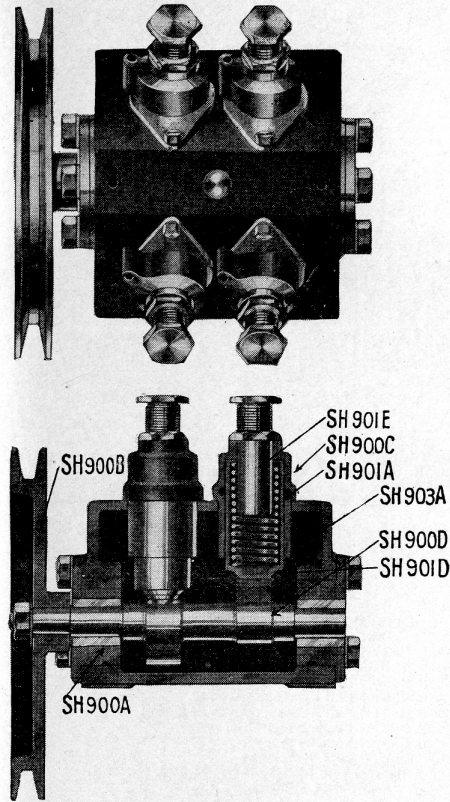
REGULATING TANK ASSEMBLY

38285—25†—13

## NOMENCLATURE LIST—Continued

| Part No.                    | Number<br>per<br>machine | Description and location                                                                |
|-----------------------------|--------------------------|-----------------------------------------------------------------------------------------|
| ENGINE—Continued            |                          |                                                                                         |
| CAM-SHAFT HOUSING—continued |                          |                                                                                         |
| 8481.....                   | 2                        | Cam-shaft bearing, No. 2 assembly.                                                      |
| 8113.....                   | 2                        | Cam-shaft bearing, upper half, No. 2.                                                   |
| 8114.....                   | 2                        | Cam-shaft bearing, lower half, No. 2.                                                   |
| 8482.....                   | 2                        | Cam-shaft bearing, No. 3 assembly.                                                      |
| 8115.....                   | 2                        | Cam-shaft bearing, upper half, No. 3.                                                   |
| 8116.....                   | 2                        | Cam-shaft bearing, lower half, No. 3.                                                   |
| 8483.....                   | 2                        | Cam-shaft bearing No. 4 assembly.                                                       |
| 8117.....                   | 2                        | Cam-shaft bearing, upper half, No. 4.                                                   |
| 8118.....                   | 2                        | Cam-shaft bearing, lower half, No. 4.                                                   |
| 8484.....                   | 2                        | Cam-shaft bearing No. 5 assembly.                                                       |
| 8119.....                   | 2                        | Cam-shaft bearing (upper half), No. 5.                                                  |
| 8120.....                   | 2                        | Cam-shaft bearing (lower half), No. 5.                                                  |
| 12485.....                  | 2                        | Cam-shaft bearing No. 6 assembly.                                                       |
| 12248.....                  | 2                        | Cam-shaft bearing (upper half), No. 6.                                                  |
| 12249.....                  | 2                        | Cam-shaft bearing (lower half), No. 6.                                                  |
| 8101.....                   | 48                       | Cam-shaft bearing upper to lower half screw (screw No. 8—32 by ½-inch special).         |
| ENGINE                      |                          |                                                                                         |
| CAM-SHAFT HOUSING           |                          |                                                                                         |
| 12250.....                  | 2                        | Cam-shaft bearing (front).                                                              |
| 8376.....                   | 2                        | Cam-shaft bearing (front) end plate.                                                    |
| 8219.....                   | 8                        | Cam-shaft bearing (front) end plate screw (screw No. 10—thd. special head)              |
| 177.....                    | 2                        | Cam-shaft bearing (front) end plate screw lock wire W. and M. Ga. No. 18 (0.0475) by 6. |
| 8379.....                   | 2                        | Cam-shaft bearing (front) end plate gasket.                                             |
| 8108.....                   | 14                       | Cam-shaft bearing lock screw.                                                           |
| 157.....                    | 14                       | Cam-shaft bearing lock screw gasket (¾-inch annular).                                   |
| 177.....                    | 12                       | Cam-shaft bearing lock screw lock wire, W. and M. Ga., No. 18 (0.0475) by 6.            |
| 8093.....                   | 2                        | Cam-shaft gear.                                                                         |
| 8059.....                   | x                        | Cam-shaft gear shim.                                                                    |
| 171.....                    | 14                       | Cam-shaft gear bolt (¼-in.—28 by ⅝-inch special) (1% inches between head and nut).      |
| 200.....                    | 14                       | Cam-shaft gear bolt nut (¼-inch—28.)                                                    |
| 106.....                    | 14                       | Cam-shaft gear bolt cotter ( ⅜ inch by ½).                                              |
| 13006.....                  | 2                        | Cam-shaft distributor driving flange.                                                   |
| 12486.....                  | 2                        | Cam-shaft housing and cover assembly.                                                   |
| 12124.....                  | 2                        | Cam-shaft housing.                                                                      |
| 8095.....                   | 10                       | Cam-shaft housing cover.                                                                |
| 8467.....                   | 2                        | Cam-shaft housing cover (propeller end).                                                |
| 8133.....                   | 2                        | Cam-shaft housing cover (propeller end plug).                                           |
| ENGINE                      |                          |                                                                                         |
| CAM-SHAFT HOUSING           |                          |                                                                                         |
| 158.....                    | 2                        | Cam-shaft housing cover (propeller end plug gasket) (¾ inch annular).                   |
| 178.....                    | 36                       | Cam-shaft housing cover bolt (⅝-inch—24 by 2 ⅜ inches special).                         |
| 102.....                    | 36                       | Cam-shaft housing cover bolt nut (⅝-inch—24 thd. standard).                             |
| 112.....                    | 72                       | Cam-shaft housing cover bolt washer (⅝-inch).                                           |
| 107.....                    | 36                       | Cam-shaft housing cover bolt cotter ( ⅜ inch by ⅝ inch).                                |
| 8088.....                   | 2                        | Cam-shaft housing front end cover.                                                      |
| 169.....                    | 2                        | Cam-shaft housing front cover gasket (1½-inch annular).                                 |
| 177.....                    | 2                        | Cam-shaft housing front end cover lock wire W. and M. Ga. No. 18 (0.0475) by 6.         |
| 8462.....                   | 2                        | Cam-shaft housing front end cover oil connection.                                       |
| 157.....                    | 4                        | Cam-shaft housing front end cover oil connection gasket (¾-inch annular).               |
| 8122.....                   | 2                        | Cam-shaft housing front end cover oil connection nut.                                   |
| 8133.....                   | 1                        | Cam-shaft housing cover (propeller end) plug.                                           |
| 8425.....                   | 12                       | Cam-shaft rocker lever, right assembly.                                                 |
| 8424.....                   | 12                       | Cam-shaft rocker lever, left assembly.                                                  |
| 8020.....                   | 12                       | Cam-shaft rocker lever (right).                                                         |
| 8019.....                   | 12                       | Cam-shaft rocker lever (left).                                                          |
| 8471.....                   | 48                       | Cam-shaft rocker lever plug (machined with lever).                                      |
| ENGINE                      |                          |                                                                                         |
| CAM-SHAFT HOUSING           |                          |                                                                                         |
| 8082.....                   | 24                       | Cam-shaft rocker lever roller.                                                          |
| 8083.....                   | 24                       | Cam-shaft rocker lever roller pin.                                                      |
| 8084.....                   | 24                       | Cam-shaft rocker lever roller pin sleeve.                                               |
| 8085.....                   | 24                       | Cam-shaft rocker lever tappet.                                                          |
| 8086.....                   | 24                       | Cam-shaft rocker lever tappet shim, medium.                                             |

Plate No. 115



AIR-PRESSURE PUMP

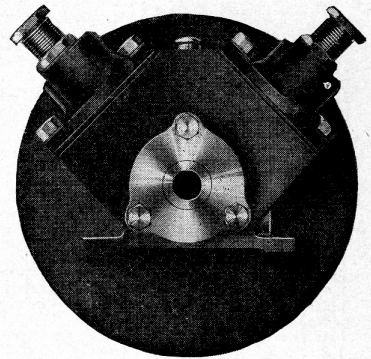
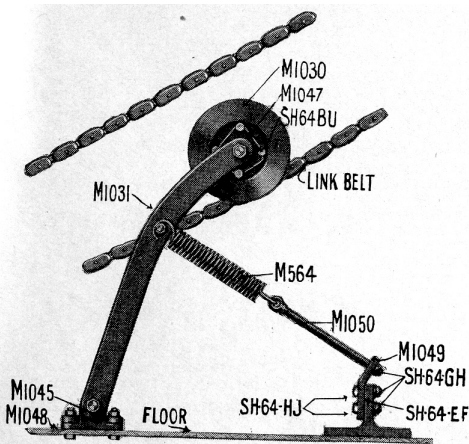
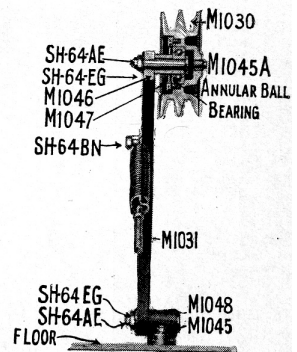


Plate No. 116



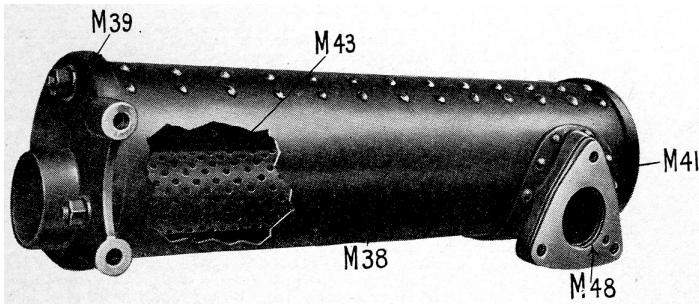
JOCKEY PULLEY



## NOMENCLATURE LIST—Continued

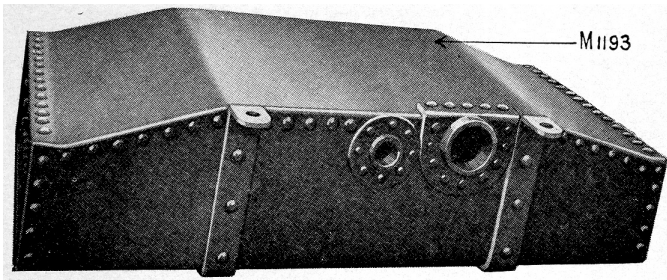
| Part No.                    | Number<br>per<br>machine | Description and location                                                                                                                                  |
|-----------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| ENGINE—Continued            |                          |                                                                                                                                                           |
| CAM-SHAFT HOUSING—continued |                          |                                                                                                                                                           |
| 8087 .....                  | 24                       | Cam-shaft rocker lever tappet shim, thin.                                                                                                                 |
| 8089 .....                  | 24                       | Cam-shaft rocker lever tappet shim, thick.                                                                                                                |
| 102 .....                   | 24                       | Cam-shaft rocker lever tappet nut.                                                                                                                        |
| 107 .....                   | 24                       | Cam-shaft rocker lever tappet cotter.                                                                                                                     |
| 8102 .....                  | 2                        | Cam-shaft driving shaft (upper).                                                                                                                          |
| 8107 .....                  | 2                        | Cam-shaft driving shaft (upper), bushing small.                                                                                                           |
| 8092 .....                  | 2                        | Cam-shaft driving shaft (upper), bushing large.                                                                                                           |
| 8094 .....                  | 2                        | Cam-shaft driving shaft (upper) gear.                                                                                                                     |
| 160 .....                   | 4                        | Cam-shaft driving shaft (upper) gear key ( $\frac{3}{8}$ by $\frac{1}{2}$ by $\frac{3}{8}$ inch).                                                         |
| 156 .....                   | 2                        | Cam-shaft driving shaft (upper) gear pin ( $\frac{1}{2}$ by $1\frac{1}{2}$ inches).                                                                       |
| 8109 .....                  | 2                        | Cam-shaft driving shaft (upper) housing and flange assem.                                                                                                 |
| 8103 .....                  | 2                        | Cam-shaft driving shaft (upper) housing.                                                                                                                  |
| 8110 .....                  | 2                        | Cam-shaft driving shaft (upper) housing flange.                                                                                                           |
| 186 .....                   | 2                        | Cam-shaft driving shaft (upper) housing brazing pin; (8d. finishing nail).                                                                                |
| ENGINE                      |                          |                                                                                                                                                           |
| CAM-SHAFT HOUSING           |                          |                                                                                                                                                           |
| 8139 .....                  | 2                        | Cam-shaft driving shaft (upper) housing flange gasket.                                                                                                    |
| 181 .....                   | 8                        | Cam-shaft driving shaft (upper) housing flange stud ( $\frac{1}{4}$ -inch—28 by $1\frac{1}{4}$ inches std.) ( $\frac{1}{4}$ inch between nut and surface) |
| 101 .....                   | 8                        | Cam-shaft driving shaft (upper) housing flange .stud nut ( $\frac{1}{4}$ -inch—28 thd., standard).                                                        |
| 106 .....                   | 8                        | Cam-shaft driving shaft (upper) housing flange stud nut ( $\frac{3}{8}$ inch by $\frac{1}{2}$ inch).                                                      |
| 8068 .....                  | 2                        | Cam-shaft driving shaft (upper) housing packing.                                                                                                          |
| 8066 .....                  | 2                        | Cam-shaft driving shaft (upper) housing packing nut.                                                                                                      |
| ENGINE                      |                          |                                                                                                                                                           |
| CAM-SHAFT LOWER SHAFT       |                          |                                                                                                                                                           |
| 8208 .....                  | 2                        | Cam-shaft driving shaft (lower) assembly.                                                                                                                 |
| 8141 .....                  | 2                        | Cam-shaft driving shaft (lower).                                                                                                                          |
| 8063 .....                  | 4                        | Cam-shaft driving shaft (lower) ball bearing.                                                                                                             |
| 8070 .....                  | 2                        | Cam-shaft driving shaft (lower) ball bearing container.                                                                                                   |
| 8343 .....                  | 2                        | Cam-shaft driving shaft (lower) ball bearing container gasket.                                                                                            |
| 8072 .....                  | 2                        | Cam-shaft driving shaft (lower) bearing container cover.                                                                                                  |
| 3144 .....                  | 2                        | Cam-shaft driving shaft (lower) bearing spacer.                                                                                                           |
| 3076 .....                  | 2                        | Cam-shaft driving shaft (lower) gear.                                                                                                                     |
| 8142 .....                  | 2                        | Cam-shaft driving-shaft (lower) nut (nut $\frac{5}{8}$ inch—18 thd. special).                                                                             |
| 172 .....                   | 2                        | Cam-shaft driving-shaft, lower cotter ( $\frac{1}{2}$ inch by $1\frac{1}{2}$ inch).                                                                       |
| 159 .....                   | 4                        | Cam-shaft driving-shaft, lower gear key ( $\frac{3}{8}$ inch by $\frac{1}{2}$ inch by $\frac{5}{8}$ inch).                                                |
| ENGINE.                     |                          |                                                                                                                                                           |
| CRANK CASE.                 |                          |                                                                                                                                                           |
| 12243 .....                 | 1                        | Crank case, lower half assembly.                                                                                                                          |
| 12036 .....                 | 1                        | Crank case, lower half.                                                                                                                                   |
| 12038 .....                 | 1                        | Crank case, lower half oil manifold.                                                                                                                      |
| 8518 .....                  | 1                        | Crank case, lower half oil-hole plug (plug $\frac{5}{8}$ inch—18 thd. Hex. head).                                                                         |
| 157 .....                   | 1                        | Crank case, lower half oil hold plug gasket ( $\frac{5}{8}$ -inch annular).                                                                               |
| 12135 .....                 | 1                        | Crank case, lower half oil tube (large).                                                                                                                  |
| 12136 .....                 | 6                        | Crank case, lower half oil tube (small).                                                                                                                  |
| 8174 .....                  | 2                        | Crank case, lower half oil pump to manifold tube.                                                                                                         |
| 12039 .....                 | 1                        | Crank case, lower half oil-pump suction tube.                                                                                                             |
| 8040 .....                  | 1                        | Crank case, lower half oil pump-suction tube extension.                                                                                                   |
| 158 .....                   | 1                        | Crank case, lower half oil-pump suction tube extension gasket ( $\frac{5}{8}$ -inch annular).                                                             |
| 161 .....                   | 1                        | Crank case, lower half oil manifold plug ( $\frac{1}{2}$ -inch pipe).                                                                                     |
| 161 .....                   | 2                        | Crank case, lower half oil-pump suction tube plug ( $\frac{1}{2}$ -inch pipe).                                                                            |
| 8033 .....                  | 14                       | Crank-shaft bearing bolt (long).                                                                                                                          |
| 8034 .....                  | 2                        | Crank-shaft bearing bolt (short).                                                                                                                         |
| 115 .....                   | 16                       | Crank-shaft bearing bolt washer $\frac{1}{2}$ (inch).                                                                                                     |
| 12548 .....                 | 16                       | Crank-shaft bearing bolt washer.                                                                                                                          |
| 110 .....                   | 32                       | Crank-shaft bearing bolt cotter ( $\frac{3}{8}$ inch by $\frac{1}{2}$ inch).                                                                              |
| 105 .....                   | 32                       | Crank-shaft bearing bolt nut ( $\frac{1}{2}$ inch—20 thd. std).                                                                                           |
| 8018 .....                  | 7                        | Crank-shaft bearing dowel.                                                                                                                                |
| ENGINE.                     |                          |                                                                                                                                                           |
| CRANK CASE                  |                          |                                                                                                                                                           |
| 132 .....                   | 10                       | Oil pump to crank-case stud ( $\frac{1}{4}$ inch—28 by $1\frac{1}{2}$ inch standard) ( $\frac{3}{8}$ inch between nut and surface).                       |
| 140 .....                   | 4                        | Water pump to crank-case stud ( $\frac{3}{8}$ inch—24 by $1\frac{1}{2}$ inch standard).                                                                   |
| 8131 .....                  | 2                        | Crank case, lower half sump (front) cover lock bushing.                                                                                                   |
| 8130 .....                  | 11                       | Crank case, lower half sump (front) cover lock.                                                                                                           |
| 8129 .....                  | 1                        | Crank case, lower half sump (front) cover.                                                                                                                |

Plate No. 117



MUFFLER

Plate No. 118.



GASOLINE TANK

## NOMENCLATURE LIST—Continued

| Part No.             | Number<br>per<br>machine | Description and Location                                                                                                                            |
|----------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| ENGINE—Continued     |                          |                                                                                                                                                     |
| CRANK CASE—continued |                          |                                                                                                                                                     |
| 8530 .....           | 1                        | Crank case, lower half sump (front) oil strainer assembly.                                                                                          |
| 8527 .....           | 1                        | Crank case, lower half sump (front) oil strainer housing.                                                                                           |
| 8528 .....           | 1                        | Crank case, lower half sump (front) oil strainer screen-side.                                                                                       |
| 8529 .....           | 1                        | Crank case, lower half sump (front) oil strainer screen-top.                                                                                        |
| 12244 .....          | 1                        | Crank case, upper half assembly.                                                                                                                    |
| 12035 .....          | 1                        | Crank case, upper half.                                                                                                                             |
| 8216 .....           | 1                        | Crank case, upper half drain tube assembly.                                                                                                         |
| 8217 .....           | 1                        | Crank case, upper half drain tube.                                                                                                                  |
| 8344 .....           | 1                        | Crank case, upper half drain tube cup.                                                                                                              |
| 8218 .....           | 1                        | Crank case, upper half drain tube flange.                                                                                                           |
| 117 .....            | 2                        | Crank case, upper half drain tube bolt ( $\frac{1}{4}$ inch—28 by $\frac{1}{2}$ inch standard) ( $\frac{1}{8}$ inch between head and nut).          |
| 101 .....            | 2                        | Crank case, upper half drain tube bolt nut ( $\frac{1}{4}$ inch—28 thd. standard).                                                                  |
| 106 .....            | 2                        | Crank case, upper half drain tube bolt nut ( $\frac{1}{8}$ inch by $\frac{1}{2}$ inch).                                                             |
| ENGINE               |                          |                                                                                                                                                     |
| CRANK CASE           |                          |                                                                                                                                                     |
| 12137 .....          | 1                        | Crank case, upper half oil tube.                                                                                                                    |
| 8156 .....           | 2                        | Crank case oil-filler assembly.                                                                                                                     |
| 8155 .....           | 2                        | Crank case oil-filler assembly.                                                                                                                     |
| 8162 .....           | 2                        | Crank case oil-filler cover assembly.                                                                                                               |
| 8151 .....           | 2                        | Crank case oil-filler cover.                                                                                                                        |
| 8157 .....           | 2                        | Crank case oil-filler cover hinge.                                                                                                                  |
| 8159 .....           | 2                        | Crank case oil-filler cover hinge pin.                                                                                                              |
| 8163 .....           | 2                        | Crank case oil-filler cover lock.                                                                                                                   |
| 8160 .....           | 2                        | Crank-case oil-filler cover spring.                                                                                                                 |
| 261 .....            | 4                        | Crank-case oil-filler cover spring escutcheon pin (No. 14 by $\frac{3}{8}$ inch).                                                                   |
| 261 .....            | 6                        | Crank-case oil-filler cover hinge escutcheon pin (No. 14 by $\frac{3}{8}$ inch).                                                                    |
| 261 .....            | 4                        | Crank-case oil-filler cover lock escutcheon pin (No. 14 by $\frac{3}{8}$ inch).                                                                     |
| 8166 .....           | 2                        | Crank-case oil-filler strainer (upper assembly).                                                                                                    |
| 8164 .....           | 2                        | Crank-case oil-filler strainer (upper ring).                                                                                                        |
| 8165 .....           | 2                        | Crank-case oil-filler strainer (upper screen).                                                                                                      |
| 8523 .....           | 2                        | Crank-case oil-filler strainer (lower assembly).                                                                                                    |
| 8522 .....           | 2                        | Crank-case oil-filler strainer (lower ring).                                                                                                        |
| 8521 .....           | 2                        | Crank-case oil-filler strainer (lower screen).                                                                                                      |
| 8158 .....           | 2                        | Crank-case oil-filler gasket.                                                                                                                       |
| 132 .....            | 4                        | Crank-case oil-filler stud ( $\frac{1}{4}$ inch—28 by $1 \frac{1}{8}$ inches, standard) ( $\frac{1}{8}$ inch between nut and surface).              |
| 111 .....            | 4                        | Crank-case oil-filler stud washer ( $\frac{1}{4}$ inch).                                                                                            |
| 101 .....            | 4                        | Crank-case oil-filler stud nut ( $\frac{1}{4}$ inch—28 thd., standard).                                                                             |
| 106 .....            | 4                        | Crank-case oil-filler stud cotter ( $\frac{1}{8}$ inch by $\frac{1}{2}$ inch).                                                                      |
| 12237 .....          | 1                        | Crank-case oil-retaining washer front end.                                                                                                          |
| 8206 .....           | 1                        | Crank-case (rear end) cover.                                                                                                                        |
| 8205 .....           | 1                        | Crank-case (rear end) cover gasket.                                                                                                                 |
| 245 .....            | 6                        | Crank-case (rear end) cover screw ( $\frac{3}{8}$ inch—24 by 1 inch).                                                                               |
| 8365 .....           | 1                        | Crank-case breather.                                                                                                                                |
| 8366 .....           | 1                        | Crank-case breather screen.                                                                                                                         |
| 8172 .....           | 1                        | Crank-case breather gasket (2-bolt gasket $\frac{1}{8}$ -inch hole).                                                                                |
| 131 .....            | 2                        | Crank-case breather stud ( $\frac{1}{4}$ inch—28 by $1 \frac{1}{4}$ inches, standard) ( $\frac{1}{4}$ inch between nut and surface).                |
| 111 .....            | 2                        | Crank-case breather stud washer ( $\frac{1}{4}$ inch).                                                                                              |
| 101 .....            | 2                        | Crank-case breather stud nut ( $\frac{1}{4}$ inch—28 thd., standard).                                                                               |
| 106 .....            | 2                        | Crank-case breather stud cotter ( $\frac{1}{8}$ inch by $\frac{1}{2}$ inch).                                                                        |
| 13148 .....          | 1                        | Crank-shaft bearing (lower half, long).                                                                                                             |
| 13418 .....          | 1                        | Crank-shaft bearing (upper half, long).                                                                                                             |
| 13459 .....          | 6                        | Crank-shaft bearing (lower half, short).                                                                                                            |
| 13435 .....          | 6                        | Crank-shaft bearing (upper half, short).                                                                                                            |
| 8390 .....           | 1                        | Propeller hub thrust bearing oil cup (brass).                                                                                                       |
| 116 .....            | 50                       | Crank-case, upper to lower half bolt (small) ( $\frac{1}{4}$ inch—28 by $\frac{1}{8}$ inch) ( $\frac{1}{2}$ inch between head and nut).             |
| 111 .....            | 100                      | Crank case, upper to lower half bolt, small washer ( $\frac{1}{4}$ inch).                                                                           |
| 101 .....            | 50                       | Crank case, upper to lower half bolt, small nut ( $\frac{1}{4}$ inch—28 thd., standard).                                                            |
| 106 .....            | 50                       | Crank case, upper to lower half bolt, small cotter ( $\frac{1}{8}$ inch by $\frac{1}{2}$ inch).                                                     |
| 187 .....            | 2                        | Crank case, upper to lower half bolt, short ( $\frac{3}{8}$ inch—24 by $\frac{1}{2}$ inches, special) ( $\frac{3}{8}$ inches between head and nut). |
| 165 .....            | 2                        | Crank case, upper to lower half bolt, long ( $\frac{3}{8}$ inch—24 by $\frac{1}{2}$ inches, special) ( $\frac{1}{2}$ inches between head and nut).  |
| 113 .....            | 8                        | Crank case, upper to lower half bolt washer ( $\frac{3}{8}$ inch).                                                                                  |
| 103 .....            | 4                        | Crank case, upper to lower half bolt nut ( $\frac{3}{8}$ inch—24 thd., standard).                                                                   |
| 108 .....            | 4                        | Crank case, upper to lower half bolt, cotter ( $\frac{1}{8}$ inch by $\frac{1}{2}$ inch).                                                           |
| 151 .....            | 100                      | Cylinder to crank-case stud ( $\frac{3}{8}$ inch—24 by $1 \frac{1}{8}$ inches, special) ( $\frac{1}{4}$ inch between nut and surface).              |
| 108 .....            | 100                      | Cylinder to crank-case stud cotter ( $\frac{1}{8}$ inch by $\frac{1}{2}$ inch).                                                                     |
| 103 .....            | 72                       | Cylinder to crank-case stud nut ( $\frac{3}{8}$ inch—24, standard).                                                                                 |
| 108 .....            | 4                        | Water pump to crank-case stud cotter ( $\frac{1}{8}$ inch by $\frac{1}{2}$ inch).                                                                   |
| 101 .....            | 3                        | Generator to crank-case stud nut.                                                                                                                   |

## NOMENCLATURE LIST—Continued

| Part No.                    | Number<br>per<br>machine | Description and location                                                                                                  |
|-----------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>ENGINE—Continued</b>     |                          |                                                                                                                           |
| <b>CRANK CASE—continued</b> |                          |                                                                                                                           |
| 103 .....                   | 4                        | Water pump to crank-case stud nut ( $\frac{3}{8}$ inch—24, standard).                                                     |
| 113 .....                   | 4                        | Water pump to crank-case stud washer ( $\frac{3}{8}$ inch).                                                               |
| 111 .....                   | 10                       | Oil pump to crank-case stud washer ( $\frac{1}{4}$ inch).                                                                 |
| 8348 .....                  | 1                        | Oil-pump body (lower half) gasket.                                                                                        |
| 106 .....                   | 10                       | Oil pump to crank-case stud cotter ( $\frac{1}{8}$ inch by $\frac{1}{2}$ inch).                                           |
| 8389 .....                  | 1                        | Generator gasket.                                                                                                         |
| 177 .....                   | 3                        | Generator to crank-case stud lock wire.                                                                                   |
| 101 .....                   | 8                        | Cam-shaft driving shaft, lower bearing connecting-stud nut ( $\frac{1}{4}$ inch—28, standard).                            |
| 106 .....                   | 8                        | Cam-shaft driving shaft, lower bearing connecting-stud cotter ( $\frac{1}{8}$ inch by $\frac{1}{2}$ inch).                |
| 12346 .....                 | 4                        | Cylinder to crank-case gasket.                                                                                            |
| 8234 .....                  | 28                       | Cylinder to crank-case stud nut.                                                                                          |
| 12353 .....                 | 1                        | Engine name plate.                                                                                                        |
| 197 .....                   | 4                        | Engine name-plate screw.                                                                                                  |
| 13157 .....                 | 1                        | Engine name plate (Navy).                                                                                                 |
| 101 .....                   | 10                       | Oil pump to crank-case stud nut.                                                                                          |
| 13443 .....                 | 10                       | Crank-shaft bearing dowel (upper).                                                                                        |
| <b>ENGINE</b>               |                          |                                                                                                                           |
| <b>CRANK SHAFT</b>          |                          |                                                                                                                           |
| 12514 .....                 | 1                        | Crank shaft, thrust bearing and gear assembly.                                                                            |
| 12546 .....                 | 1                        | Crank shaft.                                                                                                              |
| 8138 .....                  | 1                        | Crank-shaft propeller end plug.                                                                                           |
| 13142 .....                 | 6                        | Crank-shaft oil-hole plug (small).                                                                                        |
| 8067 .....                  | 1                        | Crank-shaft gear.                                                                                                         |
| 8502 .....                  | x                        | Crank-shaft gear shim.                                                                                                    |
| 198 .....                   | 6                        | Crank-shaft gear bolt ( $\frac{1}{8}$ inch—24 by $\frac{3}{8}$ inch, special) ( $\frac{1}{8}$ inch between head and nut). |
| 102 .....                   | 6                        | Crank-shaft gear bolt ( $\frac{1}{8}$ inch—24, standard).                                                                 |
| 107 .....                   | 6                        | Crank-shaft gear-bolt cotter nut ( $\frac{1}{8}$ inch by $\frac{3}{8}$ inch).                                             |
| 13481 .....                 | 1                        | Crank-shaft gear end plug.                                                                                                |
| 13482 .....                 | 1                        | Crank-shaft gear end plug gasket.                                                                                         |
| 13483 .....                 | 11                       | Crank-shaft main bearing plug.                                                                                            |
| 13484 .....                 | 12                       | Crank-pin plug.                                                                                                           |
| 13485 .....                 | 12                       | Gasket ( $1\frac{1}{4}$ inches, annular).                                                                                 |
| 13486 .....                 | 11                       | Gasket ( $1\frac{3}{8}$ inches, annular).                                                                                 |
| 13487 .....                 | 1                        | Crank-shaft gear end plug stud.                                                                                           |
| 13488 .....                 | 6                        | Crank-pin plug stud.                                                                                                      |
| 13489 .....                 | 5                        | Crank-shaft main bearing plug stud.                                                                                       |
| 103 .....                   | 12                       | Crank-pin plug-stud nut.                                                                                                  |
| 157 .....                   | 24                       | Crank-shaft plug-stud gasket.                                                                                             |
| 108 .....                   | 12                       | Crank-pin plug-stud cotter.                                                                                               |
| 13490 .....                 | 12                       | Crank-pin plug-stud nut.                                                                                                  |
| 12048 .....                 | 1                        | Propeller hub thrust bearing assembly.                                                                                    |
| 12252 .....                 | 1                        | Propeller hub thrust bearing sleeve (thick).                                                                              |
| <b>ENGINE</b>               |                          |                                                                                                                           |
| <b>CONNECTING ROD</b>       |                          |                                                                                                                           |
| 13442 .....                 | 6                        | Connecting-rod assembly.                                                                                                  |
| 13440 .....                 | 6                        | Connecting-rod (forked end) and bushings assembly.                                                                        |
| 12250 .....                 | 24                       | Connecting-rod (forked end) bolt.                                                                                         |
| 13228 .....                 | 12                       | Connecting-rod (forked end) cap.                                                                                          |
| 13220 .....                 | 24                       | Connecting-rod (forked end) bolt nut ( $\frac{1}{8}$ inch—24 thd., standard).                                             |
| 107 .....                   | 24                       | Connecting-rod (forked end) bolt cotter ( $\frac{1}{8}$ inch by $\frac{3}{8}$ inch).                                      |
| 13460 .....                 | 6                        | Connecting-rod crank-shaft bearing (upper half).                                                                          |
| 13461 .....                 | 6                        | Connecting-rod crank-shaft bearing (lower half).                                                                          |
| 8028 .....                  | 12                       | Connecting-rod crank-shaft bearing dowel.                                                                                 |
| 13441 .....                 | 6                        | Connecting-rod (plain end) and piston pin bushing assembly.                                                               |
| 13422 .....                 | 12                       | Connecting-rod (plain end) bolt.                                                                                          |
| 13227 .....                 | 6                        | Connecting-rod (plain end) cap.                                                                                           |
| 13251 .....                 | 12                       | Connecting-rod (plain end) bolt nut ( $\frac{7}{8}$ inch—20 thd., special).                                               |
| 109 .....                   | 12                       | Connecting-rod (plain end) bolt cotter ( $\frac{1}{8}$ inch by $\frac{3}{4}$ inch).                                       |
| 8007 .....                  | 12                       | Connecting-rod piston-pin bushing.                                                                                        |
| <b>ENGINE</b>               |                          |                                                                                                                           |
| <b>CYLINDER</b>             |                          |                                                                                                                           |
| 8385 .....                  | 12                       | Cylinder and valves assembly.                                                                                             |
| 8106 .....                  | 12                       | Cylinder assembly.                                                                                                        |
| 8231 .....                  | 12                       | Cylinder barrel and elbows assembly                                                                                       |
| 8098 .....                  | 12                       | Cylinder barrel.                                                                                                          |
| 8031 .....                  | 12                       | Cylinder-elbow exhaust.                                                                                                   |

## NOMENCLATURE LIST—Continued

| Part No.                    | Number<br>per<br>machine | Description and location                                                                                                                                     |
|-----------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ENGINE—Continued</b>     |                          |                                                                                                                                                              |
| <b>CYLINDER—continued</b>   |                          |                                                                                                                                                              |
| 8032.....                   | 12                       | Cylinder-elbow inlet.                                                                                                                                        |
| 8030.....                   | 24                       | Cylinder-elbow flange.                                                                                                                                       |
| 8099.....                   | 12                       | Cylinder water jacket (inlet half).                                                                                                                          |
| 8359.....                   | 12                       | Cylinder water jacket (outlet half).                                                                                                                         |
| 8025.....                   | 12                       | Valve-stem guide exhaust.                                                                                                                                    |
| 8026.....                   | 12                       | Valve-stem guide inlet.                                                                                                                                      |
| 8126.....                   | 12                       | Cylinder water-inlet pipe.                                                                                                                                   |
| 8127.....                   | 12                       | Cylinder water-outlet pipe.                                                                                                                                  |
| 8021.....                   | 24                       | Valve.                                                                                                                                                       |
| 8253.....                   | 24                       | Valve spring (inside).                                                                                                                                       |
| 8254.....                   | 12                       | Valve-spring inlet (outside).                                                                                                                                |
| 8357.....                   | 12                       | Valve-spring exhaust (outside)                                                                                                                               |
| 8090.....                   | 24                       | Valve-spring collar (lower).                                                                                                                                 |
| 8023.....                   | 24                       | Valve-spring collar (upper).                                                                                                                                 |
| 8524.....                   | 48                       | Valve-spring collar key.                                                                                                                                     |
| 8591.....                   | 24                       | Spark-plug assembly.                                                                                                                                         |
| 181.....                    | 24                       | Spark-plug gasket ( $\frac{3}{4}$ inch, annular).                                                                                                            |
| 179.....                    | 12                       | Cam-shaft housing to cylinder stud short ( $\frac{5}{16}$ inch—24 by 2 $\frac{5}{16}$ inches, special)<br>(1 $\frac{5}{16}$ inches between nut and surface). |
| 180.....                    | 12                       | Cam-shaft housing to cylinder stud long ( $\frac{5}{16}$ inch by 4 $\frac{3}{4}$ inches, special) (3 $\frac{3}{4}$<br>inches, between nut and surface).      |
| 102.....                    | 24                       | Cam-shaft housing to cylinder stud nut ( $\frac{5}{16}$ inch—24 thd., standard).                                                                             |
| 107.....                    | 24                       | Cam-shaft housing to cylinder stud cotter.                                                                                                                   |
| 162.....                    | 24                       | Exhaust header stud ( $\frac{5}{16}$ inch—24 by 1 $\frac{1}{4}$ inches, special) ( $\frac{5}{16}$ inch between nut<br>and surface).                          |
| 163.....                    | 24                       | Intake header stud ( $\frac{5}{16}$ inch—24 by 1 $\frac{1}{2}$ inches, special) ( $\frac{5}{16}$ inch between nut<br>and surface).                           |
| 103.....                    | 24                       | Intake header stud nut.                                                                                                                                      |
| 108.....                    | 24                       | Intake header stud cotter ( $\frac{5}{16}$ inch by $\frac{5}{16}$ inch).                                                                                     |
| 167.....                    | 24                       | Exhaust header stud nut ( $\frac{5}{16}$ inch—24 thd., special).                                                                                             |
| 112.....                    | 24                       | Cam-shaft housing to cylinder stud washer ( $\frac{5}{16}$ inch).                                                                                            |
| 8176.....                   | 12                       | Exhaust header gasket.                                                                                                                                       |
| 113.....                    | 24                       | Exhaust header stud washer.                                                                                                                                  |
| 108.....                    | 24                       | Exhaust header stud cotter ( $\frac{5}{16}$ inch by $\frac{5}{16}$ inch).                                                                                    |
| 113.....                    | 24                       | Intake header stud washer ( $\frac{5}{16}$ inch).                                                                                                            |
| 8173.....                   | 12                       | Intake header water inlet elbow assembly.                                                                                                                    |
| 8170.....                   | 12                       | Intake header water inlet elbow.                                                                                                                             |
| 8171.....                   | 12                       | Intake header water inlet elbow flange.                                                                                                                      |
| 8539.....                   | 12                       | Exhaust port cover DH4.                                                                                                                                      |
| 8538.....                   | 12                       | Exhaust header flange DH4.                                                                                                                                   |
| 8152.....                   | 12                       | Cylinder water hose (outlet), (see cooling system parts list).                                                                                               |
| <b>ENGINE</b>               |                          |                                                                                                                                                              |
| <b>ELECTRICAL EQUIPMENT</b> |                          |                                                                                                                                                              |
| 12229.....                  | 2                        | Distributor assembly.                                                                                                                                        |
| 8358.....                   | 2                        | Distributor to cam-shaft housing gasket.                                                                                                                     |
| 116.....                    | 10                       | Distributor bolt ( $\frac{1}{4}$ inch—28 by $\frac{5}{16}$ inches) ( $\frac{1}{2}$ inch between head and nut).                                               |
| 111.....                    | 20                       | Distributor-bolt washer ( $\frac{1}{4}$ inch).                                                                                                               |
| 101.....                    | 10                       | Distributor-bolt nut ( $\frac{1}{4}$ inch—28 std.).                                                                                                          |
| 106.....                    | 10                       | Distributor-bolt cotter ( $\frac{5}{16}$ inch by $\frac{1}{2}$ inch).                                                                                        |
| 8333.....                   | 1                        | Distributor-control connecting-rod, and clevis assembly.                                                                                                     |
| 8468.....                   | 1                        | Distributor-control connecting-rod assembly.                                                                                                                 |
| 8469.....                   | 1                        | Distributor-control connecting rod.                                                                                                                          |
| 8470.....                   | 1                        | Distributor-control connecting-rod clevis (brazing) ( $\frac{5}{16}$ -inch clevis brazing).                                                                  |
| 210.....                    | 2                        | Distributor-control connecting-rod brazing pin (2 D wire nail).                                                                                              |
| 8472.....                   | 1                        | Distributor-control connecting-rod end ( $\frac{1}{4}$ -inch rod end).                                                                                       |
| 8338.....                   | 1                        | Distributor-control connecting-rod clevis (right hand) ( $\frac{1}{4}$ inch adj. clevis<br>(right hand) thd.).                                               |
| 8340.....                   | 1                        | Distributor-control connecting-rod clevis—RH check nut ( $\frac{1}{4}$ inch, 28 thd.).                                                                       |
| 196.....                    | 2                        | Distributor-control connecting-rod clevis pin ( $\frac{1}{4}$ -inch clevis).                                                                                 |
| 106.....                    | 2                        | Distributor-control connecting-rod clevis pin cotter ( $\frac{5}{16}$ inch by $\frac{1}{2}$ inch).                                                           |
| 12490.....                  | 1                        | High-tension cable and tube assembly.                                                                                                                        |
| 12282.....                  | 1                        | High-tension cable tube.                                                                                                                                     |
| 12283.....                  | 6                        | High-tension cable-tube clip.                                                                                                                                |
| <b>ENGINE</b>               |                          |                                                                                                                                                              |
| <b>ELECTRICAL EQUIPMENT</b> |                          |                                                                                                                                                              |
| 12388.....                  | 2                        | High-tension cable-tube end.                                                                                                                                 |
| 12387.....                  | 2                        | High-tension cable separator.                                                                                                                                |
| 8332.....                   | 24                       | High-tension cable terminal (distributor end).                                                                                                               |
| 8397.....                   | 24                       | High-tension cable-terminal assembly (spark plug end).                                                                                                       |
|                             | 24                       | High-tension cable rubber sleeve (distributor end).                                                                                                          |
| 8417.....                   | 1                        | Ignition battery.                                                                                                                                            |
| 12987.....                  | 1                        | Positive battery terminal.                                                                                                                                   |
| 12988.....                  | 1                        | Negative battery terminal.                                                                                                                                   |